

Quality Regulation and Competition: Evidence from Pharmaceutical Markets*

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Abstract. Quality regulation seeks to ensure quality and foster competition by reducing vertical differentiation, but it may also have adverse effects on market structure. We study this trade-off in the context of pharmaceutical bioequivalence, which is the primary quality standard for generic drugs. By exploiting the introduction of bioequivalence in Chile, we find that stronger regulation reduced the number of drugs in the market by 18% and increased average paid prices by 13%. We estimate a model of drug entry, certification, and demand to study how drug quality, aversion to generics, and certification costs shape the equilibrium effects of quality regulation. We find that quality regulation increased demand for generic drugs by resolving asymmetric information and reducing aversion to unbranded generics, which induced the entry of high-quality drugs in place of low-quality drugs. Consumer welfare increased despite higher prices and a lower number of firms. We compare minimum quality standards with alternative designs of quality regulation, including quality disclosure.

Keywords: quality regulation, disclosure, competition, bioequivalence, generics

JEL Codes: I11, L11, L15

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1 Introduction

When consumers cannot verify product quality, the risk of purchasing a low-quality product reduces their willingness to pay. Moreover, firms' inability to signal quality harms competition, deterring entry by high-quality products and affecting their prices. Quality regulation—often in the form of minimum quality standards—is a common policy response to such market failures, particularly in credence goods markets such as pharmaceuticals. However, despite its widespread adoption, empirical evidence on the equilibrium effects of quality regulation and its mechanisms remains limited.

The effects of quality regulation are theoretically ambiguous. On the one hand, it removes products that fail to meet the minimum quality standard, thereby reducing asymmetric information and increasing perceived quality. This, in turn, encourages the entry of high-quality products and leads to lower vertical differentiation. However, costly compliance with the standard can lead to firm exit and harm price competition. Overall, the equilibrium effects of quality regulation result from the interplay between increased quality, reduced vertical differentiation, and changes in market structure driven by costly compliance.

In this paper, we study quality regulation in pharmaceutical markets, in which quality provision, asymmetric information, and quality misperceptions are central to academic and policy debates (WHO, 2000; Bate *et al.*, 2011; Bronnenberg *et al.*, 2015). Specifically, we study the impacts of bioequivalence requirements—the primary quality regulation for generic pharmaceuticals in the U.S. and Europe, and increasingly adopted elsewhere, including by China in 2016 and India in 2017. We study a bioequivalence reform in Chile that required generic drug manufacturers—i.e., producers of therapeutic copies of original innovator drugs—to provide evidence that their products are equivalent to a reference drug, usually the innovator. Products that failed to demonstrate bioequivalence by the regulatory deadline were required to exit the market. Our paper is the first to examine the impacts of this form of quality regulation on market outcomes and welfare.

Before the introduction of quality regulation, the Chilean retail pharmaceutical market displayed low generic penetration. Specifically, *unbranded generics*—those marketed under the name of the active ingredient—accounted for less than 30% of sales despite being, on average, 6 and 10 times cheaper than *branded generics*—those marketed under a proprietary name—and innovator drugs, respectively. In this context, the government introduced quality regulation to increase the perceived quality of unbranded generics and strengthen price competition.

We begin by estimating the effects of quality regulation on market outcomes. We exploit the staggered implementation of the reform, along with key features of its enforcement, to compare outcomes across and within markets (defined by the active ingredient, hereafter molecules) exposed to the policy. The regulation affected market structure by decreasing the number of drugs by 18% and increasing average paid prices by 13%; this was driven primarily by price increases

among incumbent drugs. These effects were more pronounced in smaller markets—defined as those with baseline revenue below the median—for which the number of drugs decreased by 29%, and average paid prices increased by 26%.

We develop a model to quantify the mechanisms through which quality regulation operates, assess its welfare impacts, and analyze counterfactual policies. On the demand side, consumers choose among innovators, branded generics, and unbranded generics. Demand is subject to two frictions. First, due to asymmetric information, consumers cannot distinguish between high- and low-quality generics in the absence of quality regulation. Consequently, consumers may be harmed by the presence of low-quality drugs. Second, consumers may be averse to generics, even if quality is known. Both asymmetric information and aversion reduce consumers' willingness to pay for generics relative to innovators. On the supply side, firms have private information about their quality and play a two-stage game. In the first stage, firms choose whether to participate in the market in an incomplete information game, which requires incurring an entry cost and, if quality regulation is in place, a certification cost. In the second stage, entrants compete on prices. In the model, quality regulation influences market outcomes through four mechanisms: resolving asymmetric information, removing low-quality generics from the market, altering consumer aversion to generics, and increasing fixed costs via certification requirements.

We take this model to the data to estimate its primitives. On the demand side, we recover the willingness to pay for each drug type and how it changes following the reform. From the supply side, we recover the prevalence of low-quality drugs before the reform, as well as entry costs, certification costs, and marginal costs. Our estimates imply that the regulation substantially increased the valuation of unbranded generics, with no effect on branded generics. We also find that, prior to the reform, 31% of unbranded generics and 3% of branded generics were below the quality standard. Using these estimates together, we recover consumer aversion to generics and how it changes with the regulation. We find that approximately 80% of the increase in the valuation of unbranded generics stems from resolving asymmetric information, while the remaining 20% results from reduced aversion. This decomposition clarifies how the regulation improved both the actual and perceived drug quality.

By combining the estimated model with a welfare metric that accounts for choice frictions, we find that quality regulation leads to higher welfare despite the lower product variety and higher prices. Resolving asymmetric information is the primary source of welfare gains, since it raises demand for high-quality drugs and encourages their entry. The exit of low-quality generics induced by the minimum quality standard contributes little on its own, as these products would hold a small market share if information frictions were resolved. Reduced aversion to generics lowers vertical differentiation and increases price competition, generating additional welfare gains. Finally, whereas certification costs do contribute to drug exit and higher prices, these effects are

insufficient to outweigh the welfare gains generated by the other mechanisms. Overall, welfare increases by between \$39 and \$74 million USD per year, or 3.3% and 5.5%, respectively.¹

While quality regulation increases aggregate welfare, its effects are heterogeneous across markets. In fact, welfare declines in 22% of the markets. These losses are concentrated among smaller markets, consistent with our descriptive evidence. Moreover, we find that quality regulation would generate greater welfare gains in counterfactual settings with low baseline drug quality, where resolving asymmetric information has a stronger impact.

We use the model to evaluate alternative policy designs. First, we show that certification subsidies would preserve the benefits of quality regulation while mitigating its adverse competitive effects, particularly in smaller markets. Second, we consider quality disclosure as a policy that addresses asymmetric information about product quality without restricting market entry (Dranove and Jin, 2010; Barahona *et al.*, 2023; Vatter, 2025). Compared to quality regulation, disclosure would result in less exit of unbranded generics and smaller price increases, but only marginally higher consumer welfare. However, quality regulation yields higher overall welfare by inducing less net entry and a better selection of drugs into the market. Finally, we examine policies that complement quality regulation by impacting aversion to generics. The benefits of quality regulation are enhanced when it leads to modest reductions in aversion, as this further improves decision-making and intensifies competition. However, large reductions in aversion can backfire, as they may trigger drug exit substantial enough to offset the gains from improved choice.

The main contribution of our paper is to study the equilibrium effects of quality regulation in a setting in which consumers lack information about product quality. Previous research includes a well-established theoretical literature (Leland, 1979; Shapiro, 1983; Ronnen, 1991), along with empirical work on childcare input regulation (Currie and Hotz, 2004; Hotz and Xiao, 2011); occupational licensing (Kleiner, 2000; Kleiner and Soltas, 2023; Farronato *et al.*, 2024); seller certification in e-commerce (Jin *et al.*, 2022; Hui *et al.*, 2023); and testing of medical devices (Grennan and Town, 2020). Relative to this literature, we are the first to estimate a structural model to quantify the main mechanisms at play, develop welfare analysis, and evaluate policy counterfactuals. The model is novel in combining a supply side featuring entry and price competition under quality regulation with a demand side that incorporates choice frictions that respond endogenously to the policy. This framework also allows us to empirically compare quality regulation with quality disclosure, the main alternative policy discussed in the literature (Dranove and Jin, 2010).

Second, we contribute to the literature on pharmaceutical market regulation. Most prior work

¹This range of welfare effects is derived from an analysis that accounts for the choice frictions in our demand model and considers alternative scenarios regarding the welfare-relevance of aversion to generics, which depends on the latter's underlying source. For example, aversion to generics may stem from spurious brand preferences (Bronnenberg *et al.*, 2015) or, alternatively, from genuine quality differences that persist even after bioequivalence requirements are implemented (Ishitani, 2025). Our results are qualitatively robust to these alternative assumptions for welfare analysis.

has focused on price regulation (Dubois and Lasio, 2018; Dubois and Sæthre, 2020; Dubois *et al.*, 2022; Maini and Pammolli, 2023; Cao and Gupta, 2025); off-label prescription regulation (Tunçel, 2024); public competition (Atal *et al.*, 2024); and patent protection (Chaudhuri *et al.*, 2006). To our knowledge, this is the first paper to study the equilibrium effects of quality regulation for generics.

Finally, we contribute to a literature that documents aversion to generics and, in some cases, attributes it to imperfect information, brand preferences, or actual quality differences (Ching, 2010; Bronnenberg *et al.*, 2015; Colgan *et al.*, 2015; Bairoliya *et al.*, 2017; Carrera and Villas-Boas, 2023; Cao and Gupta, 2025; Ishitani, 2025). We show how quality regulation affects information asymmetries and aversion to generics, and how those determine equilibrium outcomes. Moreover, we develop a welfare analysis that accounts for choice frictions under different assumptions regarding the welfare-relevance of aversion to generics, which may vary depending on its sources.

The paper proceeds as follows. Section 2 describes the Chilean pharmaceutical market, the policy change, and our data. Section 3 presents descriptive evidence on the effects of quality regulation on market outcomes. Section 4 provides survey evidence on perceived quality gaps across drug segments and how they relate to bioequivalence. Section 5 describes the model, and Section 6 presents its estimation. Section 7 shows our analysis of the equilibrium impacts of quality regulation, and Section 8 discusses our counterfactual policy analysis. Finally, Section 9 concludes.

2 Setting

2.1 Institutional Framework

We study the retail pharmaceutical market in Chile, which provides around 40% of prescription drugs in the country according to the 2016 National Health Survey.² In this setting, insurance coverage is limited. Approximately 80% of purchases are paid fully out-of-pocket, the exception being public insurance enrollees with a particular set of diseases (Osorio, 2020). Moreover, there is no price regulation. Hence, prices play a key role in consumer choice and drug affordability.

Three features of our setting influence the workings of the quality regulation policy. First, direct-to-consumer advertising of prescription drugs is prohibited. This could render consumers more price-sensitive, since expensive branded drugs cannot use advertising to boost demand. Second, the retail pharmacy market is highly concentrated, which may affect supply-side reactions to regulation: Three large pharmacy chains account for over 80% of the market, with the remainder shared by small local chains. Third, physician prescribing behavior and pharmacists' ability to offer alternatives to prescribed drugs influence consumer choice. In Chile, pharmacists can only offer generic substitution when prescriptions specify the generic name and a bioequivalent substitute is available. Despite recent policy efforts to constrain prescribing discretion, physicians

²The remaining 60% are provided free—or at low cost—at primary care clinics to public insurance enrollees.

still occasionally prescribe by brand name, which limits substitution toward generics.³

2.2 Quality Regulation

Bioequivalence certification establishes the therapeutic equivalence between a generic drug and a reference drug, typically the innovator drug.⁴ The Chilean government introduced bioequivalence requirements to complement existing quality regulations, which focused solely on minimum production standards and safety based on International Pharmacopeia books' guidelines (WHO, 2017), but not on therapeutic efficacy. A primary stated goal of the regulation was to increase competition and reduce prices by enhancing the perceived quality of generics.⁵

The first list of molecules subject to bioequivalence was published in 2005 by the Chilean Ministry of Health. This list included molecules covered by a major public health insurance reform known as AUGE (Bitrán *et al.*, 2010). However, it was not until 2009 that the regulator established technical standards for bioequivalence testing (Balmaceda *et al.*, 2015). Bioequivalence requirements were phased in gradually since then, covering 167 molecules as of March 2018. All new drugs containing the molecules listed in each decree were required to certify bioequivalence before obtaining a marketing license.⁶ Each decree specified the deadline for bioequivalence testing among incumbent drugs. However, enforcement occurred mostly at the time of marketing license renewal, which takes place every five years since first licensed. Drugs that lacked bioequivalence approval at the time of renewal lost their marketing license and were required to exit the market (Vasallo, 2010). We leverage this feature of enforcement for our empirical strategy below. Drugs with bioequivalence certification were allowed to remain on the market, accompanied by a mandatory label informing the consumer of their bioequivalence to the innovator.⁷

Certification deadlines were often extended due to slow uptake and capacity constraints in testing facilities. The combination of the staggered introduction of the regulation and subsequent

³In February 2014, Law 20,724 was enacted, requiring physicians to prescribe by the generic name to enable substitution with bioequivalent generics at the patient's request. However, industry actors concede that the law has not been enforced, prompting subsequent discussions in Congress.

⁴Two drugs are considered bioequivalent if the rate and extent of absorption of the tested and reference drugs do not significantly differ when administered at the same molar dose of the therapeutic ingredient under similar conditions (Davitt *et al.*, 2013). These parameters are influenced, for example, by the pharmacokinetic properties of the active and secondary ingredients, and attributes of the production process (Pramod *et al.*, 2016). Bioequivalent drugs are designed to provide the same clinical effects and safety profile as the reference drug (FDA, 2017). Consequently, bioequivalence enables regulators to rely on the pre-clinical and clinical data of the reference drug when approving a generic.

⁵For example, the Head of the National Drug Agency (ANAMED) stated: "We have no doubts that drug prices will decrease because the population will have access to a wider and more competitive drug market" (La Tercera, 2012).

⁶Bioequivalence requirements were imposed only on orally administered drugs—i.e., the requirements do not apply to topical medications, vaccines, or any other drug that is not orally administered.

⁷In practice, the label may influence demand by signaling quality. However, since drugs without bioequivalence approval are required to exit the market, the label does not convey additional information in our setting to the extent that consumers are aware of the policy. We show an example of this label in Figure A.1.

deadline extensions generates substantial policy variation.⁸ Figure 2-b summarizes this variation, which we exploit to estimate policy effects below.

Bioequivalence certification is granted after the manufacturer presents successful studies performed at private certified laboratories. For imported drugs, certifications from countries with high regulatory standards—such as Canada, the U.S., and the E.U.—are often recognized and validated in Chile.⁹ The price of bioequivalence testing at licensed laboratories is reported to range between \$48 and \$240 thousand per drug and is fully covered by the manufacturer (La Tercera, 2012; CIPER, 2015).¹⁰ To put this figure in context, the median drug in our sample had an annual revenue of \$99 thousand in 2010. Moreover, 35% and 71% of drugs had annual revenues below \$48 and \$240 thousand, respectively. Although these amounts only reflect the retail market, they suggest that the financial burden imposed by bioequivalence testing was non-negligible.

2.3 Data

We employ three data sources. First, a drug registry maintained by the Public Health Institute (ISP, 2018), which collects marketing license data for the universe of drugs (equivalent to the Orange Book in the U.S.). The registry provides information on the manufacturer, the date of a drug's first license in Chile, and the dates of subsequent license renewals. It also includes drug dosage, presentation (tablet, capsule, injectable, or other), and marketing status (prescription or over-the-counter). Second, we combine the drug registry with data on bioequivalence certification, which is also available from ISP. These data contain a list of all drugs with bioequivalence certification, including the certification date and the corresponding reference drug.

We measure market outcomes using data from IQVIA (2017) on monthly retail prices and sales between January 2010 and December 2017.¹¹ For branded drugs, the IQVIA data provide price and sales at the product level and identify each drug's laboratory, dosage, and presentation. For unbranded drugs, the data provide prices and sales at the dosage and presentation level,

⁸Among the molecules with bioequivalence requirements, there are nine unique combinations of policy dates, namely the date of the first decree, date of extensions if any, and deadlines established in the first decree and extensions. Table A.1 shows the dates of the first decree (when a requirement was announced), the last decree (when the last extension was announced), and the corresponding deadlines for each group, as well as the number of molecules. For example, Group 1 includes three molecules with their first decree in January 2011 and a deadline in February 2011. However, the original deadline was extended by a final decree in June 2013, with a deadline in December 2013.

⁹Although certification is awarded *ad eternum* for a given formula and production technology, any change requires a new certification. Generally, bioequivalence is determined through *in vivo* clinical studies for each drug presentation. However, under certain conditions, *in vitro* studies suffice for different dosages of the same drug.

¹⁰All monetary values are inflation-adjusted to 2017 prices and converted to USD at a rate of 700 CLP per USD.

¹¹IQVIA collects data from pharmacy chains and distributors. The data cover 83 local markets, which account for most of the country. To aggregate the data to the national level, we compute monthly sales by adding across local markets and monthly prices as sales-weighted averages across local markets. We adjust prices in two ways. First, we adjust for inflation using the health CPI from the National Institute of Statistics. Second, we normalize prices by calculating them per defined daily dose of the drug, allowing comparison across different presentations and dosages.

aggregated across laboratories. We focus on prescription drugs, which account for more than 90% of the drugs in the molecules we study.

We define markets at the ATC-5 (molecule) level, following Duggan *et al.* (2016).¹² We focus on 115 markets in our analysis.¹³ The resulting dataset contains 1,780 unique drugs—defined as unique combinations of drug name, dosage, and presentation—and 101 manufacturers.

2.4 Descriptive Statistics

The number of bioequivalence certifications increased throughout our study period, as shown by Figure 1-a. Certification started slowly in 2010 but increased steadily since mid-2012. By December 2017, there were 972 bioequivalent drugs, of which 631 were branded generics. The timing of certification relates to the policy roll-out. Figure 1-b displays the number of certifications around the first deadline and illustrates that (i) bioequivalence approval was rare before it became mandatory, and (ii) bioequivalence approval increased substantially after the policy change.¹⁴

The data also suggest that quality regulation affected drug entry and exit. We measure entry and exit using the ISP data on licensing and renewals. For a drug, we record an entry as obtaining a license for the first time and an exit as not renewing a license upon expiration. Figure 1-c displays the number of drugs that entered and exited the market during our sample period. Exit remained stable until late 2014 and increased thereafter. In contrast, we observe no significant changes in entry patterns over time. Figure 1-d displays the number of drugs that entered and exited over time relative to the policy and shows that the increase in exit occurred after the policy roll-out.

Regarding market outcomes, the average innovator drug is more than twice as expensive as the average branded generic, and more than seven times as expensive as the average unbranded generic. These price differences across segments reveal large premiums for innovator and branded generics within molecules before the policy change.¹⁵ Price premiums are positive and substantial

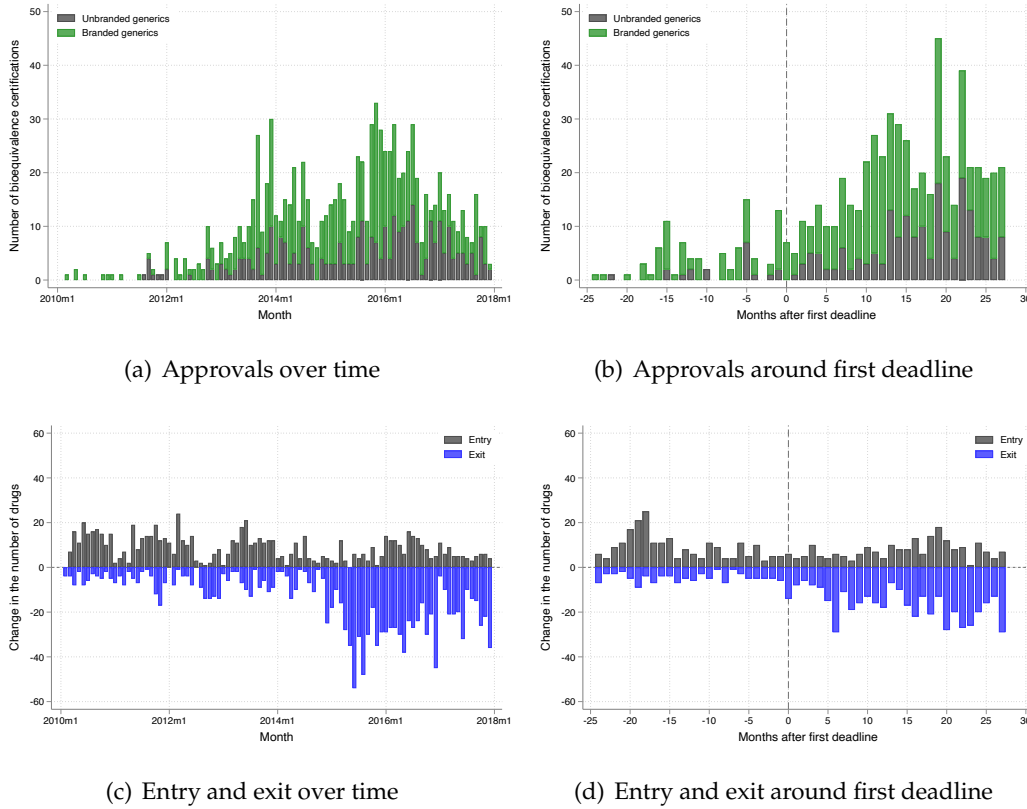
¹²This choice is motivated by the fact that the policy we study was assigned at that level. Other studies define markets at the ATC-4 level (e.g., Dubois and Lasio 2018; Dubois *et al.* 2022) to account for the potential substitution across molecules. Our results for the market effects of the policy in Section 3 are robust to using either definition.

¹³We construct our sample as follows. We begin with the 134 molecules exposed to the policy after 2010 since, as noted, technical norms for bioequivalence testing were not established until 2009. We also exclude 19 markets with sporadic sales to focus on markets that are active throughout the sample. In particular, we drop all ATC-4 groups containing a molecule for which the difference between maximum and minimum sales exceeds the 95th percentile.

¹⁴In a context in which quality is heterogeneous and unobservable to consumers, voluntary quality disclosure may occur and lead to information unraveling. In this case, consumers become aware of quality differences, and low-quality drugs may exit (Dranove and Jin, 2010). However, this prediction does not hold if disclosure is sufficiently costly (Jovanovic, 1982). Also, consumers in our setting were likely unfamiliar with the concept of bioequivalence prior to the policy change, which may have limited the returns to disclosure and helps explain the lack of voluntary disclosure.

¹⁵We estimate price premiums by regressing log real prices per defined daily dose in 2010 and 2011 on indicators for innovator and branded generics separately for each market. The exponentiated coefficients on the drug segment indicators measure the average price premiums of each segment relative to unbranded generics, which serve as the omitted category. We restrict the sample to molecules with price data for at least one innovator drug, branded drug, and unbranded drug during the period. See the distribution of price premiums in Figure A.2.

Figure 1: Quality certification, entry, and exit around policy events

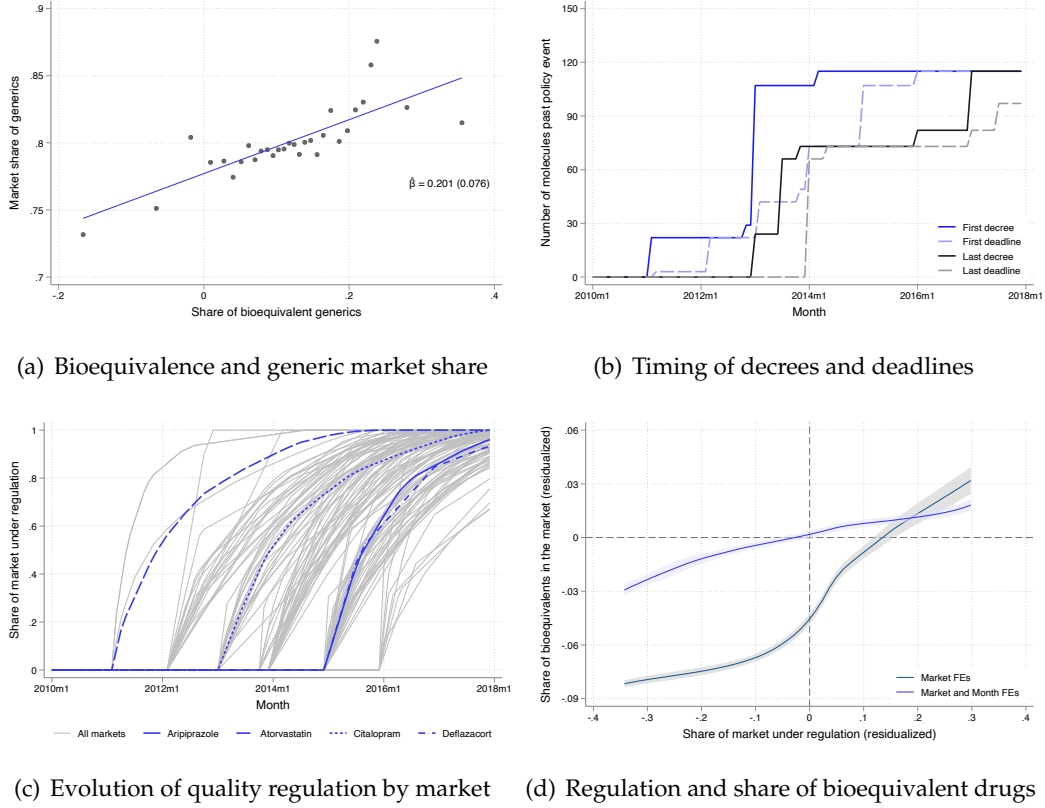


Notes: The top row displays trends in quality certification. Panel (a) displays the evolution of the number of drugs with bioequivalence approval over time, separated by unbranded (gray) and branded generics (green). Panel (b) focuses on approvals around the first bioequivalence deadline. The bottom row displays the trends in drug entry and exit. Panel (c) displays the evolution of drug entry (gray) and exit (blue) over time. Panel (d) displays entry and exit relative to the first bioequivalence deadline.

on average, with innovators and branded generics exhibiting average price premiums of 10 and 6 times relative to unbranded generics, respectively. There is considerable heterogeneity across molecules: many display price premiums of 3 to 5 times, while several exceed 10 times. See Table A.2 for additional descriptive statistics.

Innovator drugs and branded generics hold substantial market shares despite their higher prices. In 2010, branded generics had an average market share of 41%, followed by innovator drugs and unbranded generics with 34% and 25%, respectively. However, innovator drugs lost market share throughout our sample period, whereas branded and unbranded generics reached 45% and 38% average market shares by 2017. These increases coincide with a rise in the average market share of bioequivalent drugs from 0.03% in 2010 to 30% in 2017. These trends appear to be related: A regression of the market share of generics on the share of generics with bioequivalence certification within a market reveals a positive correlation, even after controlling for market and time fixed effects. In particular, an increase of 10 percentage points in the share of generics that are bioequivalent is associated with a statistically significant increase of 2 percentage points in market

Figure 2: Evolution of quality regulation



Notes: Panel (a) displays a binned scatter plot of the generic market share against the share of bioequivalent generics in a given market, controlling for market and month fixed effects. Panel (b) displays the number of markets affected by different bioequivalence policy events, from the first decree to the last deadline. Panel (c) displays the evolution over time of the treatment variable in equation (1) across markets. We highlight particular examples in blue, displayed in more detail in Figure A.3. Panel (d) displays the non-parametric relationship between the residualized policy intensity variable and the share of bioequivalent drugs in the market, controlling for market fixed effects (gray) and market and month fixed effects (blue). The bottom and top centiles of the data are not included.

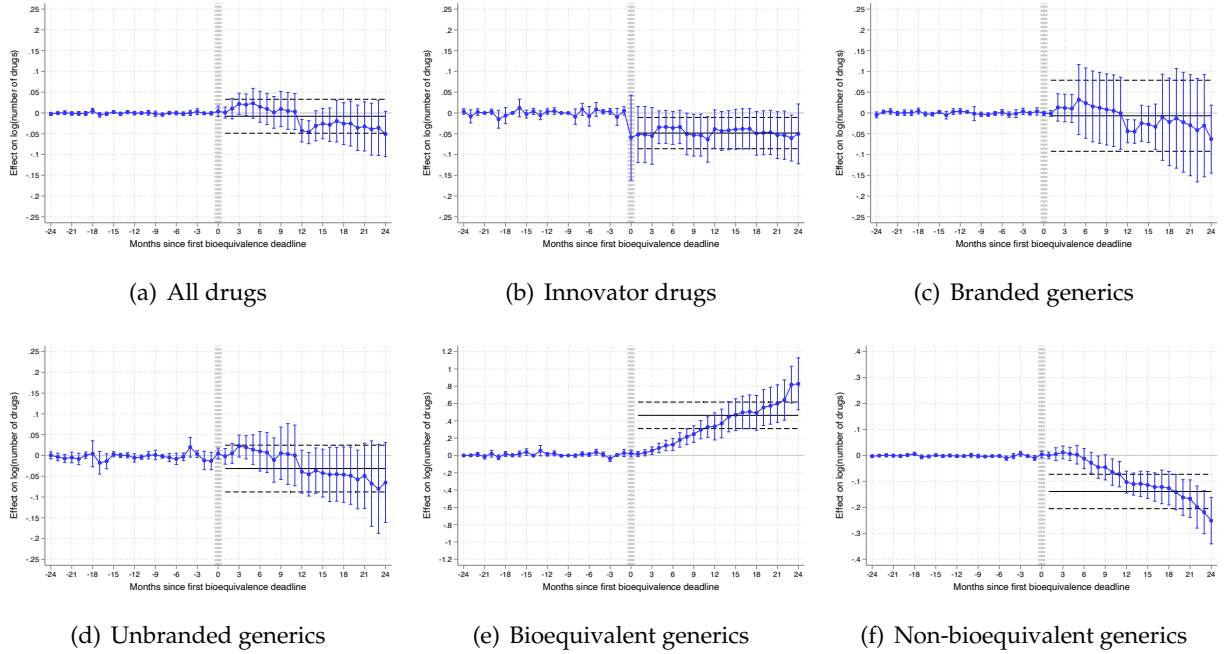
share, as shown by Figure 2-a. We further examine this relationship in the remainder of the paper, by estimating market effects in Section 3 and using a model in Section 5.

3 Effects of Quality Regulation on Market Outcomes

3.1 Event Study Evidence

We exploit the staggered roll-out of the regulation to study its effects on market outcomes. We start by implementing an event study analysis to (i) assess the assumption of parallel trends across groups of molecules treated by the policy at different times, and (ii) provide visual evidence of the effects of quality regulation on market outcomes. To accommodate potentially heterogeneous treatment effects, we employ the doubly robust difference-in-differences methods developed by Sant’Anna and Zhao (2020) and Callaway and Sant’Anna (2021). We treat the first bioequivalence deadline as the policy event and report results covering the two years before and after this event.

Figure 3: Event study results for market structure



Notes: This figure displays estimates from the event study specifications for market structure outcomes, based on the methods in Sant’Anna and Zhao (2020) and Callaway and Sant’Anna (2021). The specification uses the first deadline as the policy event and includes market and time fixed effects. Dots indicate point estimates and lines indicate 95% confidence intervals based on standard errors clustered at the market level. Coefficients are displayed for 24 months before and 24 months after the policy event. Solid horizontal lines indicate the estimated average treatment effect. Dashed horizontal lines represent the 95% confidence interval for the average treatment effect. The coefficient for the month prior to the event is normalized to zero.

The results suggest that the policy strongly affected market structure, as shown by Figure 3. Our estimates indicate that the total number of drugs decreased, which seems to be largely driven by the exit of unbranded generics. The results also show a large increase (decrease) in the number of bioequivalent (non-bioequivalent) generics after the policy change. Similarly, Figure 4 displays results related to drug prices, market shares, and sales.¹⁶ We find that prices increased after the policy change, particularly among unbranded generics. In terms of quantities, we find no clear evidence of effects on total sales or market shares by segment. We provide a more detailed discussion of the effects along all these margins in Section 3.2 below.

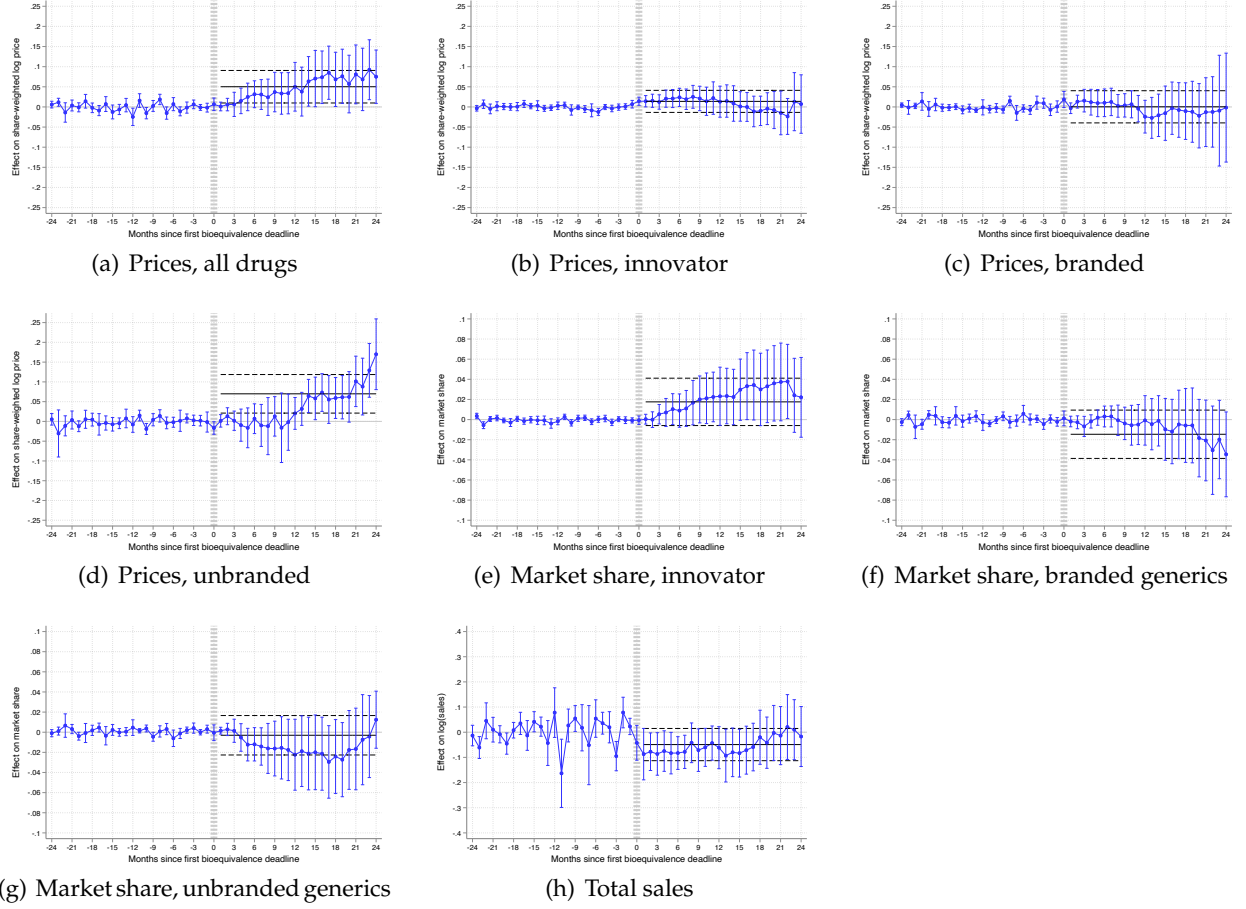
Overall, the event studies suggest that stronger quality regulation decreased the number of drugs in the market and increased prices. Additionally, the trends in outcomes before the first bioequivalence deadline are well-behaved, which validates the use of the differential timing of

¹⁶We use a price index constructed as the share-weighted average of log prices in a market (e.g., Chevalier *et al.* 2003):

$$\hat{P}_{mt} \equiv \sum_{i \in \mathcal{I}_{mt}} w_{it} P_{it},$$

where $\mathcal{I}_{mt}$ is the set of drugs in market m in period t , P_{it} is the logarithm of price per defined daily dose of drug i in period t , and w_{it} denotes the share of sales of drug i in market m in period t .

Figure 4: Event study results for prices, sales, and market shares



Notes: This figure displays estimates from the event study specifications for price and quantity outcomes, based on the methods in Sant’Anna and Zhao (2020) and Callaway and Sant’Anna (2021). The specification uses the first deadline as policy event and includes market and time fixed effects. Dots indicate point estimates and lines indicate 95% confidence intervals based on standard errors clustered at the market level. Coefficients are displayed for 24 months before and 24 months after the policy event. Solid horizontal lines indicate the estimated average treatment effect. Dashed horizontal lines represent the 95% confidence interval for the average treatment effect. The coefficient for the month prior to the event is normalized to zero.

bioequivalence requirements across markets as identifying variation.

3.2 Regression Analysis

Our main empirical strategy exploits policy variation across and within markets. The first source of variation is the staggered roll-out of the reform. The second comes from a particular institutional feature: in practice, deadlines for incumbent drugs become binding when a drug is required to renew its marketing license, which occurs every five years since first licensed. At that point, the ISP denies license renewal for drugs lacking bioequivalence approval (Vasallo, 2010). Thus, for each drug, the first license renewal following the policy deadline constitutes the effective deadline. For drugs registered before the policy announcement, renewal dates are plausibly exogenous with

respect to the policy. Moreover, renewal dates vary across drugs within a market due to baseline variation in licensing dates. Consequently, the share of drugs for which the policy effectively binds varies both across markets—despite a common policy deadline—and within markets over time.

We combine these sources of variation into a variable that measures the evolution of the policy roll-out in each market. This variable captures three key features of the policy. First, the policy becomes relevant for a market only after its first decree. Second, the policy becomes increasingly relevant for each drug as its license renewal date approaches. Finally, the policy is fully implemented in a market when the license renewal date has passed for all drugs. Formally, denote the policy date for market m by t_m^d and the renewal date of drug i in m by t_{im}^r . For a drug i , the share of time between the decree and the next renewal date that has passed by time t is

$$T_{imt} = \begin{cases} 0 & \text{if } t \leq t_m^d \\ \frac{t - t_m^d}{t_{im}^r - t_m^d} & \text{if } t_m^d < t \leq t_{im}^r \\ 1 & \text{if } t_{im}^r < t \end{cases}$$

For each market m , we define the *share of market under regulation* by month t as the average of T_{imt} across the set of generic drugs in market m during the baseline period $t_m^d, \mathcal{G}_m$, which is

$$T_{mt} = \frac{1}{|\mathcal{G}_m|} \sum_{i \in \mathcal{G}_m} T_{imt}, \quad (1)$$

where $|\mathcal{G}_m|$ is the number of branded and unbranded generics in market m at month t_m^d . Our analysis uses T_{mt} as the treatment variable. T_{mt} is a weakly increasing function of time relative to the policy date t_m^d : It equals zero before t_m^d and reaches one after the latest renewal date among drugs in $\mathcal{G}_m$. Figure 2-c displays the evolution of T_{mt} over time for all markets, highlighting substantial variation both across markets and within markets over time.¹⁷ Finally, Figure 2-d shows that this variable is indeed correlated with the share of bioequivalent drugs in the market.

Our main specification for estimating policy effects on market-level outcomes y_{mt} is

$$y_{mt} = \beta T_{mt} + \theta_m + \delta_t + \varepsilon_{mt}, \quad (2)$$

where the coefficient of interest is β ; θ_m are market fixed effects that control for permanent differences across markets; and δ_t are month fixed effects that control for common time shocks across markets. When discussing results, we focus on the impact of moving from not being subject to

¹⁷For illustration, Figure A.3 shows examples of the evolution of T_{mt} over time for four markets, along with the evolution in the number of bioequivalent drugs. These plots show how bioequivalence certification increases as bioequivalence requirements become relevant for a market. We highlight these examples in Figure 2-c.

regulation to being entirely subject to it, which is captured by increasing T_{mt} from zero to one.

The key identifying assumption is that there are no unobserved market-specific trends driving both the timing of the policy roll-out and the outcomes of interest. This assumption requires that policy deadlines and renewal dates were not set as a function of unobserved shocks not captured by market and time fixed effects. A violation of this assumption would occur if, for example, decrees and deadlines were set earlier for markets expected to have earlier price increases. We cannot directly test this assumption, but the fact that decree extensions were mostly set based on the capacity constraints of testing facilities makes it unlikely that they were driven by unobserved future demand or supply shocks. Moreover, market-level observables do not exhibit a clear correlation with the policy timing, lending support to this assumption.¹⁸ Finally, evidence of parallel pre-trends from the event study analysis further supports our identifying assumption.

We focus on market size as a dimension for heterogeneity analysis. This exercise is motivated by the intuition that when compliance is costly, quality regulation should have stronger effects among smaller markets because it would induce more drug exit. We test this prediction by estimating differential effects by market size. Specifically, we divide markets according to whether their total market revenue in 2010 was above or below the median.

Effects on market structure. We start by estimating equation (2) for the number of drugs in the market using a Poisson regression (Chen and Roth, 2023). Column 1 in Table 1-A shows that the policy decreased the total number of drugs by 18%. Columns 2–8 split this result across drug segments. The exit of branded and unbranded generics drives this overall reduction. While the number of bioequivalent generics increases, it does not compensate for the exit of non-bioequivalent generics. We find no significant changes in the number of innovator drugs.

Adverse effects on the number of drugs are larger in smaller markets, as shown by Table 1-B. We estimate that the number of drugs decreased by 29% in smaller markets and 12% in large markets, a decline primarily driven by the exit of branded and unbranded generics.

Effects on drug prices. The price effects of quality regulation are theoretically ambiguous: Whereas a reduction in the number of firms may weaken competition and lead to price increases, changes in perceived quality may reduce vertical differentiation and thereby enhance competition. We find that average prices across all drugs increased by 13% due to the regulation, as shown in Table 2-A. This increase in average paid prices is driven by increases among unbranded generics, whereas we find no statistically significant effects for innovators and branded generics.¹⁹

¹⁸Table A.1-B presents statistics for market outcomes in 2010 for markets affected by the policy at different points in time. While there is substantial heterogeneity across these groups in terms of the number of drugs, market size, and market outcomes, no clear pattern emerges that is related to the timing of the policy implementation.

¹⁹We construct the same price index by drug segment, but define the weights as within-segment shares. We estimate these regressions for the subset of markets that had at least one drug in that segment during the baseline period.

Table 1: Effects of quality regulation on number of drugs

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Dep. var.: Number of Drugs							
	All	Innovator	Branded generics			Unbranded generics		
			All	BE	Non-BE	All	BE	Non-BE
<i>A - Average effects</i>								
Regulation	-0.20*** (0.05)	-0.02 (0.06)	-0.21*** (0.05)	1.55*** (0.26)	-0.38*** (0.06)	-0.20*** (0.06)	2.78*** (0.42)	-0.37*** (0.06)
<i>B - Heterogeneity by market size</i>								
Regulation × Low Revenue	-0.34*** (0.07)	-0.12 (0.09)	-0.30*** (0.08)	1.42*** (0.30)	-0.31*** (0.09)	-0.40*** (0.09)	3.29*** (0.70)	-0.28*** (0.09)
Regulation × High Revenue	-0.13** (0.05)	0.03 (0.07)	-0.17*** (0.06)	1.62*** (0.34)	-0.40*** (0.06)	-0.08 (0.08)	2.58*** (0.43)	-0.42*** (0.08)
Pre-regulation average	22.06	2.40	12.32	0.09	12.23	7.34	0.01	7.33
Observations	11,040	9,600	10,272	8,064	10,272	10,560	7,008	10,560
Market FE	Y	Y	Y	Y	Y	Y	Y	Y
Month FE	Y	Y	Y	Y	Y	Y	Y	Y

Notes: Each column is a Poisson regression of the number of drugs in a segment on the policy roll-out variable constructed using the first decree deadline. Panel B provides results by baseline revenue. Markets are classified as low or high revenue based on the total market revenue in 2010 relative to the median across markets in that year. The sample size in column 1 is the number of markets (115) by the number of months (96). Columns 2–8 omit markets with no variation in the number of products in the segment. Standard errors clustered at the market level are reported in parentheses. ***p<0.01, **p<0.05, *p<0.1.

As shown above, the decrease in the number of drugs is concentrated among smaller markets, and hence these are the markets in which we expect to find the strongest price effects—the results in Table 2-B confirm this. The overall increase in drug prices is largely driven by a 26% increase in smaller markets, which is also concentrated among unbranded generics.²⁰

Effects on market shares and sales. Fewer generics on the market, combined with their price increases, may shift consumption away from generics. However, changes in the perceived quality of generics may increase their demand. On net, we find limited effects of quality regulation on segment market shares, as shown by columns 5–9 in Table 2-A. If anything, we estimate a statistically significant decrease in the market share of branded generics, with slight increases in the market share of innovators. As expected, we find a significant increase in the market share of bioequivalent generics and a decrease in the share of non-bioequivalent generics. Regarding heterogeneity, unbranded generics decreased their market share due to quality regulation in smaller markets, where adverse effects on the number of firms and prices were stronger.

Theoretically, quality regulation can increase or decrease the market share of the outside option

²⁰In Appendix A, we develop a decomposition of price effects into changes in individual drug prices and changes in market shares, which we further decompose into changes in market shares among incumbents, entrants, and exiters. We find that price increases by incumbent drugs account for roughly two-thirds of the overall rise in prices, consistent with both weaker competition resulting from exit and a higher willingness to pay for these drugs after the policy, as it reveals their high quality. Perhaps for this latter reason, relatively high prices set by new entrants account for about one-third of the overall price increase. Changes in valuations constitute a key component of our model in Section 5.

Table 2: Effects of quality regulation on drug prices and quantities

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)		
	Drug price index ($\hat{P}_{mt}$)				Quantity outcomes							
	All	Innovator	Branded generics	Unbranded generics	Innovator	Market shares			Unbranded generics	log(Sales)		
						Branded generics				All	BE	Non-BE
<i>A - Average effects</i>												
Regulation	0.12** (0.05)	0.04 (0.03)	-0.01 (0.05)	0.16*** (0.05)	0.08** (0.03)	-0.07** (0.03)	0.12*** (0.03)	-0.18*** (0.04)	-0.01 (0.03)	-0.10 (0.08)		
R ²	0.99	0.99	0.99	0.99	0.89	0.94	0.55	0.86	0.96	0.97		
<i>B - Heterogeneity by market size</i>												
Regulation × Low revenue	0.23*** (0.08)	0.05 (0.04)	0.00 (0.05)	0.16** (0.07)	0.11** (0.04)	-0.04 (0.03)	0.06 (0.05)	-0.10** (0.05)	-0.07* (0.04)	-0.15 (0.12)		
Regulation × High revenue	0.05 (0.06)	0.03 (0.03)	-0.01 (0.05)	0.17** (0.07)	0.06 (0.03)	-0.09** (0.04)	0.16*** (0.04)	-0.24*** (0.04)	0.03 (0.03)	-0.06 (0.09)		
R ²	0.99	0.99	0.99	0.99	0.89	0.94	0.55	0.86	0.96	0.97		
Observations	11,040	8,141	8,707	5,238	11,040	11,040	11,040	11,040	11,040	11,040		
Market FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y		
Month-Segment FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y		

Notes: Columns 1-4 display results from regressions of share-weighted log prices by molecule on the policy roll-out variable constructed using the first decree deadline. Columns 5-9 display results from regressions of market shares on the policy roll-out variable. Column 10 displays a regression of total log sales on the policy roll-out variable. Panel B provides results by baseline market size. Markets are classified as low or high revenue based on the total market revenue in 2010 relative to the median across markets in that year. The sample size in column 1 is the number of markets (115) by the number of months (96). The sample size in columns 2–4 is the number of market-month observations with at least one product in the segment. We drop markets without products in the segment at baseline (January 2010). Standard errors clustered at the market level are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

due to the interplay between changes in market structure, price effects, and perceived quality. We do not find statistically significant effects of quality regulation on total drug sales, as shown by column 10 in Table 2.

Summary of the descriptive evidence. Stronger quality regulation may have reduced vertical differentiation, increased willingness to pay for generics, and enhanced price competition. However, our results suggest that these positive effects were at least partially overturned by the adverse competitive effects of drug exit. Most of the adverse effects of stronger regulation on drug exit and higher prices are concentrated among smaller markets. This pattern suggests that drugs exit when the certification cost is large relative to market profitability.^{21,22}

While lower variety and higher prices often suggest welfare decreases, that implication is not

²¹In Appendix B, we provide evidence that neither drug recalls nor adverse health events associated with drug consumption increased after the reform. Although these results are informative, these quality metrics mostly capture drug safety rather than drug efficacy. Hence, they are not directly associated with bioequivalence, so we interpret this result cautiously. Using our model, we indirectly infer the improvement in drug quality due to bioequivalence.

²²In Appendix C, we expand the heterogeneity analysis by classifying drugs by whether they treat a chronic condition and whether they have high mortality impacts. We find stronger effects on prices and quantities among markets for non-chronic conditions, which is consistent with information frictions playing a role in the effects of the reform, as captured by our model in Section 6. We also find large price effects among drugs with high mortality impacts, which suggests potentially large welfare impacts. We compute the overall welfare effect of the reform in Section 7.2.

immediate in this context. If the distribution of drug quality improved due to the regulation, then consumers may be better off despite lower variety and higher prices. The model we develop and estimate in Section 5 allows us to quantify the welfare effects of quality regulation, unpack its equilibrium effects, and study alternative policy designs.

4 Evidence on Perceived Quality from a Consumer Survey

Our findings suggest that quality regulation had unintended consequences. The estimated price increases indicate that the adverse competitive effects of drug exit outweighed the potential pro-competitive effects from reduced vertical differentiation. However, the welfare effects of the regulation depend critically on whether consumers accurately perceive drug quality at baseline and how regulation may impact those perceptions. To explore this, we administered a consumer survey to shed light on quality perceptions and the role of bioequivalence in shaping them.

We conducted in-person surveys with frequent consumers recruited outside pharmacies. We focused on Atorvastatin, a common anti-cholesterol drug. We asked consumers about their quality perceptions over four products: the innovator drug (Lipitor, by Pfizer), a bioequivalent branded generic (Lipoten, by Pharmavita), and bioequivalent and non-bioequivalent unbranded generics (Atorvastatina, by Mintlab). We surveyed 401 consumers, of which 58% reported having a household member with a chronic disease and 34% reported buying Atorvastatin for a household member. See Appendix D for more details and Table A.5 for summary statistics.

Consumers display substantial variation in their perceptions of drug quality. We collect data on perceived quality using a numerical scale and compute the innovator's perceived quality premium relative to each of the three generic alternatives. On average, the innovator is rated as having a higher perceived quality than each of these alternatives.²³ Also, consumers perceive branded generics to be of higher quality than unbranded generics, which suggests that consumers attach a quality premium to brands. Finally, consumers perceive bioequivalent drugs to be of higher quality than non-bioequivalent drugs. Consumers thus attribute a quality premium to bioequivalence that is large enough to close the brand premium but not the innovator's perceived quality premium. These insights guide our demand model in Section 5, in which drug valuations may vary by segment and with bioequivalence certification. Moreover, we note that other factors—such as biased beliefs regarding quality, lack of understanding about bioequivalence, or detailing—may also affect perceived valuations. We refer to these factors as aversion to generics in our model.

²³Figure A.4 displays the distribution of perceived quality premiums. Relative to innovator drugs, perceived quality is 10% ($0.62\sigma_I$) lower for branded generics, 11% ($0.68\sigma_I$) lower for unbranded bioequivalent generics, and 26% ($1.62\sigma_I$) lower for unbranded non-bioequivalent generics, where σ_I is the standard deviation of the innovator's perceived quality. The p -value of a test of difference in means is zero for each of the three perceived quality premiums.

5 A Model of Entry, Certification, and Demand

In this section, we develop a model that captures the primary channels through which quality regulation operates: resolving asymmetric information, eliminating low-quality drugs, shifting aversion to generics, and imposing costly certification. Our goals are to quantify the contribution of these mechanisms to the effects of quality regulation on market outcomes documented in Section 3, to assess the welfare impacts of this regulation, and to evaluate alternative policy designs.

5.1 Environment

Market structure and drug quality. There is a set of drugs $\mathcal{P}_m$ that may enter market m in period t . Each drug $j \in \mathcal{P}_m$ belongs to a segment $k \in \{I, B, U\}$: innovators (I), branded generics (B), or unbranded generics (U). Moreover, each drug has a binary quality level $\psi_j \in \{\psi_L, \psi_H\}$ drawn from a segment-specific Bernoulli distribution with parameter π_H^k measuring the share of high-quality drugs in segment k . We define high quality as being bioequivalent and normalize ψ_L and ψ_H to 0 and 1, respectively. Innovators are bioequivalent by definition, so $\pi_H^I = 1$.

Timing. We consider two environments: one without quality regulation and one with regulation. In each environment, market participation and price competition unfold as a two-stage game. In the first stage, potential entrants $\mathcal{P}_m$ decide whether to enter based on expected profits net of fixed costs. Firms observe all attributes relevant to demand and costs for all potential entrants, but quality and profit shocks are private information, with common knowledge of their distributions. This incomplete information entry game determines the market structure $\mathcal{J}_{mt}$ (Aradillas-López, 2020). After entry decisions are made, demand and marginal cost shocks are realized. In the second stage, firms set prices in a Bertrand-Nash equilibrium, and demand is realized.

Demand. Consumers choose among drugs by trading off perceived quality, price, and other attributes. In each market m and period t , consumer i selects a product j in segment k or the outside option $j = 0$ of not purchasing. The indirect utility that consumer i derives from drug j is

$$u_{ijmt}^k = v_{jmt}^k + h(y_{it}, p_{jmt}) + x'_{jmt}\beta + \xi_{jmt} + \zeta_{imt}^k + (1 - \sigma)\epsilon_{ijmt}, \quad (3)$$

where v_{jmt}^k is the perceived value of the drug, $h(\cdot)$ is a function of consumer income y_{it} and drug price p_{jmt} , x_{jmt} are observable product characteristics, and ξ_{jmt} is a product-specific unobservable. The term $\zeta_{imt}^k + (1 - \sigma)\epsilon_{ijmt}$ captures idiosyncratic preferences, with σ controlling the degree of correlation across products within segment k (Berry, 1994). A higher value of σ implies stronger substitution within segments, intensifying business-stealing effects and the potential for excessive entry (Berry *et al.*, 2016).

A key object of interest is the perceived value of a drug, v_{jmt}^k . We consider two frictions that create a gap between the perceived value of generics and that of the innovator. First, asymmetric

information regarding the quality of generics. Consumers cannot discern the quality of a drug j , and form rational expectations based on its segment k and bioequivalence certification status, $\mathbf{b}_{jmt}$. We write the expected quality of generics as $E[\psi_j | k, \mathbf{b}_{jmt}]$. Second, consumers may exhibit aversion to generics, even if certified as bioequivalent. This aversion could arise from several sources, such as biased beliefs over quality, differences in other dimensions of quality (e.g., side effects), detailing, and others. We do not attempt to distinguish these and rather let $\tau_m^k(\mathbf{b}_{jmt})$ capture aversion to generics broadly. For welfare analysis, we consider alternative assumptions regarding the welfare relevance of aversion to generics.

We define the perceived value of a drug as a function of these frictions as

$$v_{jmt}^k \equiv \underbrace{\mu_m^I}_{\text{Valuation of drug quality}} \cdot \underbrace{(E[\psi_j | k, \mathbf{b}_{jmt}])}_{\text{Expected drug quality}} - \underbrace{\tau_m^k(\mathbf{b}_{jmt})}_{\text{Generic aversion in } k}, \quad (4)$$

where the term μ_m^I is the perceived value of an innovator in market m relative to the outside option. Note that both lower expected quality and higher aversion to generics increase perceived vertical differentiation between innovator drugs and generics in segment k .

The model captures how quality regulation affects demand by resolving asymmetric information and impacting aversion to generics. With respect to asymmetric information, bioequivalence serves as a certain signal of high quality, since only high-quality drugs can obtain certification. Therefore, the expected quality of a drug j without quality certification is $E[\psi_j | k, \mathbf{b}_{jmt} = 0] = \pi_H^k$, whereas for a drug with certification it becomes $E[\psi_j | k, \mathbf{b}_{jmt} = 1] = 1$. In addition, we let aversion to generics $\tau_m^k(\mathbf{b}_{jmt})$ vary by segment and certification status. We adopt the shorthand notation $\tau_{0,m}^k \equiv \tau_m^k(0)$ and $\tau_{1,m}^k \equiv \tau_m^k(1)$ for the levels of aversion to generics in segment k and market m before and after bioequivalence certification, respectively. Moreover, we impose by assumption that there is no aversion to innovator drugs, such that $\tau_m^I = 0$.

Under this structure, the perceived value of a drug j depends only on its segment and bioequivalence certification. As shorthand notation, we denote the perceived value of drug j in market m before and after bioequivalence certification by $v_{0,m}^k$ and $v_{1,m}^k$, respectively. For example, $v_m^I = \mu_m^I$ is the perceived value of the innovator, $v_{0,m}^B = \mu_m^I(\pi_H^B - \tau_{0,m}^B)$ is the perceived value of branded generics in market m without bioequivalence certification, and $v_{1,m}^U = \mu_m^I(1 - \tau_{1,m}^U)$ is the perceived value of unbranded generics in market m with bioequivalence certification.

Supply. We model firms' participation and pricing decisions in each market as a two-stage game. In the first stage, potential entrants in $\mathcal{P}_m$ simultaneously decide whether to enter market m in period t . In the second stage, entrants $\mathcal{J}_{mt} \subseteq \mathcal{P}_m$ maximize profits by competing on prices. Each potential entrant j draws quality ψ_j , drug attributes x_{jmt} , marginal cost c_{jmt} , and a fixed cost χ_{jmt} —which includes an entry cost F_{jm} for new entrants and a certification cost κ_{jm} when quality

regulation is in place. We assume that marginal costs and fixed costs are unrelated to quality within a segment. This assumption allows for differences in these costs across segments but implies that high- and low-quality drugs within a segment share the same systematic components.

Upon entry, the variable profits of firm j under market structure $\mathcal{J}_{mt}$ are given by

$$\tilde{\Pi}_{jmt}(\mathcal{J}_{mt}) = \max_{p_{jmt}} (p_{jmt} - c_{jmt}) q_{jmt}(\mathbf{p}_{mt}; \mathcal{J}_{mt}),$$

where $q_{jmt}(\mathbf{p}_{mt}; \mathcal{J}_{mt})$ denotes the demand for firm j given prices $\mathbf{p}_{mt}$ set by all firms in $\mathcal{J}_{mt}$. If the firm enters, it also gets an idiosyncratic profit shock ε_{jmt}^1 , whereas if it does not enter, it gets an idiosyncratic profit shock ε_{jmt}^0 from an outside option.

Firms hold incomplete information about their rivals. Firm attributes, marginal costs, entry costs, and certification costs, $\{x_{jmt}, c_{jmt}, F_{jm}, \kappa_{jm}\}$, as well as the set of potential entrants $\mathcal{P}_m$ are common knowledge among all potential entrants. In contrast, firms know only the distributions of profit shocks $(\varepsilon_{jmt}^1, \varepsilon_{jmt}^0)$ and product quality ψ_j (i.e., they know π_H^k), while the realizations of these variables are private information.

Quality regulation. Under quality regulation, branded and unbranded generics can only participate in the market if they incur a cost κ_{jm} to test for bioequivalence. Regulation may affect outcomes through three channels. First, only high-quality firms can stay in the market, and hence expected quality becomes $E[\psi_j | k, b_{jmt} = 1] = \psi_H \forall k$. Second, the regulation may affect perceived drug valuations by reducing aversion to generics, such that $\tau_m^k(b_{jmt})$ may adjust. Finally, certification costs may affect the equilibrium market structure and prices.

5.2 Equilibrium

An equilibrium without quality regulation is such that all firms in the market make non-negative expected profits net of entry costs while maximizing profits given the realized market structure. An equilibrium with quality regulation is defined similarly, but expected profits are net of entry and certification costs for new entrants and net of certification costs for incumbent firms.

Because of private information, firm j does not know the realized market structure $\mathcal{J}_{mt}$ when it considers entering the market. Let $\mathcal{S}_m$ be the set of potential market structures and $\mathcal{J}_s$ be an element of $\mathcal{S}_m$. Then, the expected profits net of fixed costs χ_{jmt} are

$$E_{\phi_{mt}}[\Pi_{jmt}(\mathcal{J})] = \underbrace{\sum_{\mathcal{J}_s \in \mathcal{S}_m} \tilde{\Pi}_{jmt}(\mathcal{J}_s) P_{jmt}^S(\mathcal{J}_s; \phi_{mt}) - \chi_{jmt} + \varepsilon_{jmt}^1}_{\equiv E_{\phi_{mt}}[\tilde{\Pi}_{jmt}(\mathcal{J})]},$$

where the expectation is taken over the entry probabilities of all potential entrants, ϕ_{mt} , $P_{jmt}^S(\mathcal{J}_s; \phi_{mt})$ is the probability that firm j assigns to market structure $\mathcal{J}_s$ given entry probabilities ϕ_{mt} , and χ_{jmt}

includes entry and possibly certification costs. This probability $P_{jmt}^S(\mathcal{J}_s; \phi_{mt})$ takes the form

$$P_{jmt}^S(\mathcal{J}_s; \phi_{mt}) = \frac{1}{\phi_{jmt}} \prod_{l \in \mathcal{J}_s, l \in \mathcal{P}_m} \phi_{lmt} \prod_{h \notin \mathcal{J}_s, h \in \mathcal{P}_m} (1 - \phi_{hmt}). \quad (5)$$

A firm enters whenever the expected profits from entering exceed those from not entering. Therefore, the probability that firm j enters the market is

$$\phi_{jmt} = \Pr(E_{\phi_{mt}}[\tilde{\Pi}_j(\mathcal{J})] - \chi_{jmt} + \varepsilon_{jmt}^1 > \varepsilon_{jmt}^0).$$

Equilibrium without quality regulation. In the absence of quality regulation, an equilibrium market structure is such that all entrants make non-negative expected profits and all non-entrants make negative expected profits, given a fixed cost $\chi_{jmt} = F_{jm}$.

Let ϕ_{mt}^{NQ} be the vector of equilibrium entry probabilities for all potential entrants, which induces probabilities $P_{jmt}^S(\mathcal{J}_s; \phi_{mt}^{NQ})$ in equation (5). Then, the expected total profits for firm j are

$$E_{\phi_{mt}^{NQ}}[\Pi_{jmt}(\mathcal{J})] = E_{\phi_{mt}^{NQ}}[\tilde{\Pi}_{jmt}(\mathcal{J})] - F_{jm} + \varepsilon_{jmt}^1 \quad \forall j \in \mathcal{P}_m,$$

which implies entry probabilities

$$\phi_{jmt}^{NQ} = \Pr(E_{\phi_{mt}^{NQ}}[\tilde{\Pi}_{jmt}(\mathcal{J})] - F_{jm} + \varepsilon_{jmt}^1 > \varepsilon_{jmt}^0) \quad \forall j \in \mathcal{P}_m, \quad (6)$$

which is a system of equations on ϕ_{mt}^{NQ} that describes the equilibrium of the entry model, with one equation for each of the P_m potential entrants.

Equilibrium with quality regulation. The basic structure of equilibrium changes along two dimensions in the environment with quality regulation. First, entry costs change due to the addition of certification costs. In particular, innovators only incur a fixed cost to enter, and thus $\chi_{jmt} = F_{jm}$ for $k_j = I$. In contrast, branded and unbranded generics incur both fixed and certification costs to enter, and thus $\chi_{jmt} = F_{jm} + \kappa_{jm}$ for $k_j \in \{B, U\}$. Second, entry probabilities change due to the selection of low-quality drugs out of the market. We define entry probabilities under quality regulation as $\phi_{jmt|L}$ and $\phi_{jmt|H}$ for low- and high-quality drugs, respectively.

Under quality regulation, the probability that firm j assigns to market structure $\mathcal{J}_s$ depends on the likelihood that branded and unbranded generics are of high quality, π_H^k , and is given by

$$P_{jmt}^S(\mathcal{J}_s; \phi_{mt}, \pi_H) = \frac{1}{\phi_{jmt|H} \pi_H^k} \prod_{l \in \mathcal{J}_s, l \in \mathcal{P}_m} \phi_{lmt|H} \pi_H^k \prod_{h \notin \mathcal{J}_s, h \in \mathcal{P}_m} (1 - \phi_{hmt|H} \pi_H^k) \quad \forall j \in \mathcal{P}_m,$$

which reflects the fact that only high-quality firms remain in the market under quality regulation.

Expected profits are calculated using these probabilities.

The equilibrium market structure under quality regulation is characterized by the following set of equations governing entry probabilities

$$\begin{aligned}\phi_{jmt|L}^Q &= 0 & \forall j \in \mathcal{P}_m, \psi_j &= \psi_L \\ \phi_{jmt|H}^Q &= \Pr(E_{\phi_{mt}^Q} [\widetilde{\Pi}_{jmt}(\mathcal{J})] - \chi_{jmt} + \varepsilon_{jmt}^1 > \varepsilon_{jmt}^0) & \forall j \in \mathcal{P}_m, \psi_j &= \psi_H,\end{aligned}\tag{7}$$

which is a system of P_m equations on ϕ_{mt}^Q .

Pricing. In the second stage of the game, the set of entrants compete by setting prices in a Nash-Bertrand game. Equilibrium prices are the solution to the following first-order conditions:

$$p_{jmt} = c_{jmt} - \frac{q_{jmt}(\mathbf{p}_{mt})}{\frac{\partial q_{jmt}(\mathbf{p}_{mt})}{\partial p_{jmt}}} \quad \forall j \in \mathcal{J}_{mt},\tag{8}$$

such that firms charge a markup over marginal cost that reflects consumer price sensitivity.

5.3 Model Discussion

While our model captures key mechanisms at play, it also abstracts from certain aspects. In this section, we discuss some of our model assumptions and how they may influence the interpretation of our results.

Exogenous drug quality. We assume that drug quality is exogenous, thereby limiting the extent to which firms could react to regulation by upgrading quality. In terms of timing, this assumption implies that regulation occurs after production technology is set up. Although this is a reasonable assumption in the short run, the long-run effects of quality regulation may include endogenous responses along this margin or, more broadly, in terms of R&D by pharmaceutical firms.

The role of physicians. We do not formally model the role of physicians in consumer decision-making. Given that we do not observe prescriptions in our data, we cannot disentangle such influence. Both misalignment between physicians and consumers and barriers to generic substitution will load on our estimates of aversion to generics.²⁴

Risk aversion. Risk aversion could play a role in drug choice and partly explain aversion to generics. However, we note that our results in Section 6.3 below are mostly inconsistent with aversion to the risk of a drug not being bioequivalent. First, we estimate substantial aversion to both branded and unbranded generics, even though the average quality of the former is close

²⁴A recent report by the Chilean antitrust authority (FNE, 2019) shows that retail pharmacies pay, on average, 70% more than the public sector, and 60% more than institutional private buyers (primarily, private hospitals) for the same drug. It links this price gap with individuals having stronger brand preferences than institutional buyers.

to that of innovator drugs. Second, aversion to unbranded generics remains high even after the reform, when uncertainty about bioequivalence is fully resolved. Since we cannot rule out aversion to risk along other quality dimensions, we consider welfare metrics that impose different assumptions about how aversion to generics affects welfare.

Dynamics and belief updating. We abstract from consumer learning about drug quality. Instead, consumers form rational expectations based on the drug’s segment and certification status, without updating their beliefs based on personal experience or the timing of certification. Since quality discovery is more likely to occur through physicians (Crawford and Shum, 2005), learning should play a limited role for unbranded generics, as prescriptions cannot specify a particular manufacturer. We note that branding of generics may serve as a quality signal, consistent with our findings that branded generics are more often high-quality and face higher entry costs in Section 6.2. We also rule out inference based on certification timing—e.g., that early certification signals quality. In practice, certification overwhelmingly takes place at the time of the first marketing license renewal, suggesting that the strategic value of early certification is limited.

Supply-side dynamics. We model entry and certification as a static game of market participation, separately for before and after the regulation. This simplifies the analysis while capturing key institutional features, such as the differential costs faced by incumbents and new entrants. Although firms could in principle observe their competitors’ certification choices, we do not model sequential decisions. The data suggest limited scope for strategic first-mover advantages: certification is concentrated at the earliest mandated renewal date, largely driven by regulation rather than timing incentives. While a richer dynamic model could allow for studying strategic preemption or learning across firms, the observed behavior and institutional setting support our approach as a reasonable approximation.

6 Econometric Model

6.1 Demand Estimation

A key object of interest in the demand model is the perceived valuation of a drug, v_{jmt}^k , which we parameterize for estimation as

$$v_{jmt}^k = v_m^I - \Delta_m^k + \eta^k b_{jmt}, \quad (9)$$

where v_m^I , Δ_m^k and η^k are parameters to be estimated. This parameterization allows the value of innovators and generics to be market-specific through the terms v_m^I and $v_m^I - \Delta_m^k$, which are market-segment fixed effects. The parameter Δ_m^k captures the gap in valuation between the innovator and the generic in segment k before bioequivalence certification. Finally, bioequivalence certification reduces the gap between the valuation of generics and innovators by a segment-specific factor η^k .

These parameters map to the structural parameters of demand in equation (4) as follows:

$$v_m^I = \mu_m^I, \quad \Delta_m^k = \mu_m^I(1 - \pi_H^k + \tau_{0,m}^k), \quad \eta^k = \mu_m^I(1 - \pi_H^k + \tau_{0,m}^k - \tau_{1,m}^k), \quad (10)$$

such that v_m^I is equal to the valuation of innovator drugs, and Δ_m^k and η^k are functions of parameters that govern aversion to generics τ_m^k and asymmetric information π_H^k . Specifically, the gap in valuation between innovators and generics before bioequivalence approval, Δ_m^k , is due to a combination of quality differences captured by $1 - \pi_H^k$ and aversion to generics $\tau_{0,m}^k$. Changes in the valuation of generics, η^k , stem from the update of expected quality captured by $1 - \pi_H^k$ and changes in aversion to generics induced by the policy, $\tau_{0,m}^k - \tau_{1,m}^k$.

To estimate the demand model, we specialize the utility function in equation (3) by specifying the price-income component as $h(y_{it}, p_{jmt}) = \frac{\alpha}{\varphi_{mt}} \ln y_{it} - \alpha \ln p_{jmt}$, which corresponds to the constant-expenditure model described by Björnerstedt and Verboven (2016).^{25,26} In this framework, consumers allocate a fixed share φ_{mt} of their income y_{it} to the pharmaceutical market and spend it entirely on their preferred alternative.²⁷ This specification yields the estimating equation

$$\ln s_{jmt} - \ln s_{0mt} = v_m^I - \Delta_m^k + \eta^k b_{jmt} - \alpha \ln p_{jmt} + x'_{jmt} \beta + \lambda_t + \sigma \ln s_{jmt|k} + \xi_{jmt}, \quad (11)$$

where the shares are expenditure shares—rather than quantity shares as in standard unit-demand models (Berry, 1994; Björnerstedt and Verboven, 2016). The expenditure share of a drug j is $s_{jmt} \equiv p_{jmt} q_{jmt} / B_{mt}$, where B_{mt} is the total market budget allocated to the pharmaceutical category, including the outside option.²⁸ The dependent variable is the difference in log expenditure shares between drug j and the outside option. The term λ_t captures time fixed effects, $s_{jmt|k}$ is the within-segment expenditure share of product j , and ξ_{jmt} is an unobserved demand shock.

We use annual data on market shares, prices, and drug attributes to estimate equation (11). We define a drug as a combination of ATC-5, segment, and producer. Although a firm can offer

²⁵This specification builds on the discrete-continuous demand in Hanemann (1984). Aggregate demand arises from individual utility-maximizing behavior. Consumers choose the product that yields the highest utility from the choice set $\mathcal{J}_{mt}$. Conditional on this choice, the quantity purchased follows from Roy's identity, $-(\partial u_{ijmt}^k / \partial p_{jmt}) / (\partial u_{ijmt}^k / \partial y_{it})$. Demand for product j , denoted $q_{jmt}(\mathbf{p}_{mt}; \mathcal{J}_{mt})$, aggregates individual quantities over the distribution of consumers and depends on the prices and characteristics of all products in the market. Appendix E.1 provides further detail.

²⁶A log price specification fits our data better than including price in levels due to the large heterogeneity in prices across markets. Related work has also adopted this specification for the same reasons (Dubois *et al.*, 2022). In addition, we note that having prices in logs in a standard discrete choice model with (assumed) unit demand is not an economically consistent model and, therefore, does not allow for a welfare analysis consistent with utility maximization. In contrast, the constant-expenditure model we employ allows for both having prices in logs and a well-founded welfare analysis.

²⁷This yields realistic heterogeneity in drug purchases across income—especially in a setting with substantial out-of-pocket expenditure like Chile. In Appendix E.1, we use survey data to verify that drug spending is approximately proportional to income. Note that although income does not impact choice probabilities, it matters for welfare analysis.

²⁸We do not observe B_{mt} but instead recover it using a calibration approach, following recent work estimating drug demand using similar data (e.g., Dubois and Lasio 2018). See Appendix E.2 for details.

each drug in multiple presentations, we treat all of them as a single product and include the number of presentations as a product attribute, as in Dubois and Lasio (2018). For each product, we measure prices p_{jmt} as the volume-weighted average price per defined daily dose of each drug, and b_{jmt} as the share of presentations with bioequivalence certification; we allow for heterogeneous effects of this variable on demand for branded and unbranded generics. Besides the number of presentations, drug attributes x_{jmt} include the share of presentations that have been on the market for five years or more, which serves as a measure of product age. Finally, we include year fixed effects in λ_t to capture changes in the relative attractiveness of the outside option over time.²⁹

Identification. The main identification concern in equation (11) is that pricing and certification decisions may be driven by unobserved preference shocks ξ_{jmt} . In addition, the conditional market share $s_{jmt|k}$ is correlated with ξ_{jmt} by construction. We employ instrumental variables to obtain consistent estimates and use GMM. We instrument prices using average prices in Norway within the corresponding ATC-5 category (Norwegian Institute of Public Health, 2019). Norwegian prices are regulated based on a basket of prices from nine European countries (Brekke *et al.*, 2015) and capture overall changes in production costs by ATC-5 category over time. We instrument bioequivalence certification using the share of presentations past their market's certification deadline and their marketing license renewal deadline, calculated separately for branded and unbranded generics. Finally, we instrument within-segment shares using the average age of competing drugs, which captures variation in the relative appeal of alternatives within a segment.

Results. We summarize our demand estimates in Table 3-A. We estimate substantially higher price elasticities for generics than for innovators. Own price elasticities are, on average, 2.4 for innovators, 4.6 for branded generics, and 4.9 for unbranded generics, which are in line with recent estimates in the literature (e.g., Dubois and Lasio, 2018). Moreover, our nesting parameter estimate $\hat{\sigma} = 0.57$ indicates substantially stronger substitution within a segment than across segments.³⁰

To describe the baseline gaps in perceived drug valuation between segments, we fit our demand model to an environment prior to quality regulation, $b_{jmt} = 0$, to obtain estimates of perceived quality valuation for each market and segment, $\hat{v}_{0,m}^k$. Before the regulation, the average perceived valuation of innovator drugs was 2.3 times that of unbranded generics, which helps explain their low market share despite lower prices. In contrast, the average perceived valuation of innovator drugs was only 1.3 times that of branded generics, consistent with the latter's high market share despite higher prices.

²⁹ A limitation of the sales data from IQVIA is that unbranded generics are reported as a bundle, with no information on individual laboratories. We describe how we address this issue in Appendix E.3.

³⁰ Table A.3 shows auxiliary linear regressions of each endogenous variable on the instruments separately. As expected, our instrument for prices based on prices in Norway is associated with higher drug prices in Chile. Moreover, license renewals have strong effects on drug certification, consistent with the evidence presented earlier.

Table 3: Model estimates

	(1)	(2)	(3)	(4)	(5)
<i>A - Demand side estimates</i>	GMM				
log Price (α)	1.81 (0.71)				
Bio, branded (η^B)	-0.30 (0.33)				
Bio, unbranded (η^U)	1.23 (0.50)				
Nesting parameter (σ)	0.57 (0.10)				
Observations	4,011				
<i>B - Supply side estimates</i>	Marginal cost (γ_c)	Entry cost (\$1,000s)	Certification cost (\$1,000s)	Profit shock (γ_{σ_e})	Drug quality (π_H)
Innovator		0.00 (0.00)			1.00 -
Branded	-0.24 (0.02)	435.57 (223.44)			0.97 (0.01)
Unbranded	-1.49 (0.03)	340.44 (34.57)			0.69 (0.12)
Developing country origin			181.89 (44.71)		
Developed country origin			0.00 (0.00)		
log(Market size)				0.35 (0.03)	
Constant				8.42 (0.42)	
Observations	6,572	4,508	4,508	4,508	4,508
<i>C - Aversion to generics</i>	Before regulation (τ_0^k)		After regulation (τ_1^k)		
	Mean	SD	Mean	SD	
Branded	0.25 (0.04)	0.43	0.38 (0.04)	0.43	
Unbranded	0.98 (0.09)	0.91	0.91 (0.09)	0.89	

Notes: Panel A shows demand estimates. The specification includes market and year fixed effects. Panel B shows supply model estimates. Marginal cost is estimated on a panel of yearly observations for 2010–2017. The specification includes market and year fixed effects. The remaining parameters are estimated on the cross-sections of 2010 and 2017. Entry and certification costs are reported in USD thousands, and the distribution of drug quality is reported as the probability that a drug is of high quality. Panel C shows the mean and standard deviation of estimated aversion to generics before and after the regulation. Standard errors in parentheses.

Quality regulation affected demand differently across segments. Our estimate of η^U is large and positive, indicating that the relative valuation of unbranded generics increased following the reform. In particular, the average valuation of innovator drugs under regulation is 1.9 times that of unbranded generics, as shown by Figure 5. These increases in valuations help rationalize the finding from Section 3 that unbranded generic market shares did not decrease after the reform despite their price increases. In contrast, our estimate of η^B is not significantly different from zero and negative, such that, if anything, the relative valuation of branded generics decreased after the reform. This slight decrease helps rationalize the fact that branded generic market shares decreased after the reform, despite their prices not increasing.

Together, these results suggest that quality regulation decreased vertical differentiation across generic segments, thereby intensifying competition among them. Note that valuations depend on both expected quality and aversion to generics. From equation (4), it is easy to see that we cannot disentangle the contribution of each based on demand-side estimates alone. However, recovering the share of high-quality drugs from the supply side allows us to make such a decomposition.

6.2 Supply Estimation

The objects of interest are marginal costs c_{jmt} , entry costs F_{jm} , certification costs κ_{jm} , the probability that a drug is of high quality π_{Hj} , and the scale of profit shocks σ_ε . We parameterize them as

$$\begin{aligned} c_{jmt} &= \exp(x'_{c,jmt}\gamma_c + \omega_{jmt}) \\ F_{jm} &= \exp(x'_{F,jm}\gamma_F) \\ \kappa_{jm} &= \exp(x'_{\kappa,jm}\gamma_\kappa) \\ \pi_{Hj} &= \Lambda(x'_{\pi,jm}\gamma_\pi) \\ \sigma_{\varepsilon j} &= \exp(x'_{\sigma_\varepsilon,jm}\gamma_{\sigma_\varepsilon}), \end{aligned}$$

such that the parameters of interest are γ_c , γ_F , γ_κ , γ_π , and $\gamma_{\sigma_\varepsilon}$. The covariates included in the specification vary across equations. Marginal costs are specified as a combination of a vector of observables $x_{c,jmt}$ that includes indicators for drug segments, markets, and years, as well as cost shocks ω_{jmt} . Entry costs are allowed to vary across drug segments to capture potential investments by branded drugs in establishing and marketing their brand. Certification costs are allowed to differ depending on whether the drug was manufactured in a developing or developed country, which is motivated by the fact that the latter were granted waivers, as discussed in Section 2.2. The probability of being of high quality is specified as constant within each segment, and Λ is the logistic function, which ensures that $\pi_{Hj} \in [0, 1]$. Finally, the profit shocks $(\varepsilon^1_{jmt}, \varepsilon^0_{jmt})$ are iid T1EV with scale σ_ε , which we specify as a function of a constant and the log of market size.

We begin by estimating marginal costs. Using our demand estimates, we invert the pricing first-order condition in equation (8) to obtain estimates for marginal costs for each product-year observation, $\hat{c}_{jmt}$. Assuming that cost shocks ω_{jmt} are mean independent of the observable determinants of costs, we estimate γ_c from a linear regression of $\log \hat{c}_{jmt}$ on $x_{c,jmt}$.

We estimate the remaining supply-side parameters using firm entry choices before and after quality regulation. As discussed above, the equilibrium in our model is described by a fixed point whereby the probability that a firm enters is a function of equilibrium entry probabilities of all other potential entrants, which motivates a nested fixed-point likelihood estimator (e.g., Seim 2006). Such an approach is computationally intensive, as it requires solving for an equilibrium for each candidate set of parameters for each market-year—this includes solving the second-stage

pricing game at each iteration. To avoid this issue at the estimation stage, we instead adopt a two-step estimator (Sweeting, 2009; Bajari *et al.*, 2010).

In the first step, we estimate entry probabilities directly from the data. Specifically, we estimate logit models of entry and certification choices on all their determinants as predicted by the model—namely own and rival drivers of variable profits, fixed costs, and certification costs, as well as market fixed effects—to obtain $\hat{\phi}_{jmt}^{NQ}$ and $\hat{\phi}_{jmt}^Q$. We then simulate variable profits $\tilde{\Pi}_{jmt}(\mathcal{J})$ under all potential market structures, using our estimates of preferences and marginal costs. Solving for optimal prices and implied demand yields variable profits for each potential entrant and potential market structure. Combining these with fitted entry probabilities from the previous step, we compute expected variable profits by integrating over potential market structures. We assume that demand and cost shocks are realized after entry and certification choices. In particular, we assume that ξ_{jmt} and ω_{jmt} are iid conditional on the information sets held by firms when deciding about entry and certification. This assumption rules out selection into the market by potential entrants based on knowledge of those unobservables.

In the second step, we use the estimated expected variable profits and the equilibrium conditions on entry probabilities in equations (6) and (7), to recover entry costs, certification costs, and the share of high-quality generic drugs among potential entrants.³¹ The distributional assumption on $(\varepsilon_{jmt}^1, \varepsilon_{jmt}^0)$ yields entry probabilities with the usual logit form, which are

$$\begin{aligned}\phi_{jmt}^{NQ} &= \frac{\exp(\frac{1}{\sigma_\varepsilon} [E_{\hat{\phi}_{jmt}^{NQ}} [\tilde{\Pi}_{jmt}(\mathcal{J})] - x'_{F,jm} \gamma_F])}{1 + \exp(\frac{1}{\sigma_\varepsilon} [E_{\hat{\phi}_{jmt}^{NQ}} [\tilde{\Pi}_{jmt}(\mathcal{J})] - x'_{F,jm} \gamma_F])} \\ \phi_{jmt}^Q &= \frac{\exp(\frac{1}{\sigma_\varepsilon} [E_{\hat{\phi}_{jmt}^Q} [\tilde{\Pi}_{jmt}(\mathcal{J})] - x'_{F,jm} \gamma_F - (1 - d_j^I) x'_{\kappa,jm} \gamma_\kappa])}{1 + \exp(\frac{1}{\sigma_\varepsilon} [E_{\hat{\phi}_{jmt}^Q} [\tilde{\Pi}_{jmt}(\mathcal{J})] - x'_{F,jm} \gamma_F - (1 - d_j^I) x'_{\kappa,jm} \gamma_\kappa])} \left[\frac{\exp(x'_{\pi,jm} \gamma_\pi)}{1 + \exp(x'_{\pi,jm} \gamma_\pi)} \right]^{1-d_j^I},\end{aligned}$$

for the environments without and with quality regulation, and where d_j^I indicates that j is an innovator drug. We use these probabilities to estimate the remaining parameters by maximum likelihood. We provide further details in Appendix E.5.

We estimate the model using two cross sections—2010 and 2017—to capture the pre- and post-reform periods, respectively. To account for incumbency, we assume that firms active before the reform do not incur the entry cost but must pay the certification cost to remain in the market after the policy change. In addition, the set of potential entrants in each market includes all firms that participated in the market at any point during the sample period, as well as the five firms with the highest participation in each segment nationally.

³¹To put variable profits, entry costs, and certification costs on the same scale, we use an annual discount rate of 0.05 to compute the present value of a stream of variable profits.

Identification. The model identification follows the framework for static entry games with incomplete information from Aradillas-López (2020). Our model is guaranteed to have an equilibrium, although it may not be unique. To achieve point identification, we assume that the equilibrium selection mechanism is degenerate, which implies that the data is generated from a single equilibrium without needing to specify which equilibrium is selected.^{32,33} This is a common assumption in the literature (Seim, 2006; Pesendorfer and Schmidt-Dengler, 2008; Bajari *et al.*, 2010; Bhattacharya *et al.*, 2025).

We exploit variation in expected variable profits to identify the parameters governing market participation. Entry depends on expected variable profits as well as entry costs F , certification costs κ , the share of high-quality drugs π_H^k , and the scale of profit shocks σ_ε . Expected variable profits vary across firms and markets—due to differences in competition, product attributes, and costs—and we can compute them based on our estimates of demand, marginal costs, and entry probabilities. We rely on this variation to identify the remaining structural parameters from how observed entry rates respond to profitability across firms, markets, and regulatory regimes.

Having data for the periods before and after the regulation aids with identification. Before the regulation, only entry costs F and the scale of profit shocks σ_ε are relevant for entry choices, and hence these parameters can be identified from the pre-regulation data alone. Segment-specific entry costs are identified from observed entry rates conditional on expected variable profits by segment. The scale of profit shocks is identified from the sensitivity of entry rates to expected profits.

After the regulation, firms also face a certification cost κ and, in addition, products below the quality standard must exit. Since F and σ_ε are identified from pre-regulation entry behavior, we discuss the identification of the certification cost κ and the share of high-quality drugs π_H^k conditional on those parameters. Both a higher κ and a lower π_H^k induce net exit in the model. However, a higher κ leads to higher exit among marginally profitable firms, whereas a lower π_H^k induces exit even among highly profitable firms. Hence, excess exit among highly profitable firms identifies π_H^k , and additional exit among less profitable firms identifies κ .³⁴ Heterogeneity in κ by

³²Unlike previous work studying settings in which firms play a discrete game repeatedly within a market (e.g., Sweeting 2009), we study two such instances for a cross-section of many markets. Since we control for a rich set of drivers of expected profits and market fixed effects when estimating entry probabilities, this assumption requires that the same equilibrium is played across markets, up to observable differences across firms and unobservable time-invariant differences across markets.

³³Under this assumption, the possibility of multiple equilibria does not affect estimation, as the data are assumed to arise from one particular equilibrium, and the two-step estimator recovers the corresponding entry probabilities. However, multiplicity can pose challenges for counterfactual simulations, where alternative equilibria might arise. To assess this, we check whether our computed equilibrium entry probabilities are sensitive to different starting values, as done in related previous work (e.g., Seim 2006). We find no evidence of multiple equilibria in our application.

³⁴This pattern holds under standard distributional assumptions on profit shocks, including the T1EV we adopt. For illustration, Figure A.5 displays the empirical relationship between post-regulation exit rates and baseline revenues.

country of origin is identified by differences in net exit between drugs of different origin, conditional on their profitability. Similarly, heterogeneity in π_H^k by segment is identified by differences in excess exit among highly profitable firms across segments.

Results. Our supply-side estimates are displayed in Table 3-B. The marginal cost estimates imply average markups of 21% for unbranded generics, 22% for branded generics, and 44% for innovators. We estimate an average entry cost of \$370 thousand. In our model, this cost is not interpreted as the cost of developing a drug but rather the cost of marketing it. Entry costs vary across segments, with branded generics incurring entry costs 28% higher than unbranded generics; this may reflect investments the former incur in establishing their brand. In contrast, we estimate that innovator drugs face essentially no entry costs. This is somewhat mechanical, since only one firm in each market is an innovator and is almost always active. Finally, we estimate certification costs of \$182 thousand for drugs from developing countries—which falls within the range reported in Section 2.2—and essentially zero for drugs from developed countries, as expected. To put these numbers in context, our estimates imply that the average generic earns annual variable profits of \$46 thousand and that 46% of generics had variable profits below the annualized certification cost prior to the policy change.

Baseline quality is high, but not all drugs are of high quality. In particular, the shares of high-quality branded and unbranded generics before the reform are $\hat{\pi}_H^B = 0.97$ and $\hat{\pi}_H^U = 0.69$, respectively. These results are consistent with some of the drug exit we documented earlier being driven by the lower average quality of generics compared to innovators. As we discuss below, this implies that quality regulation impacts the pool of drugs in the market and shifts the perceived quality of generics upward by resolving asymmetric information about drug quality.

6.3 Disentangling Aversion to Generics and Quality Differences

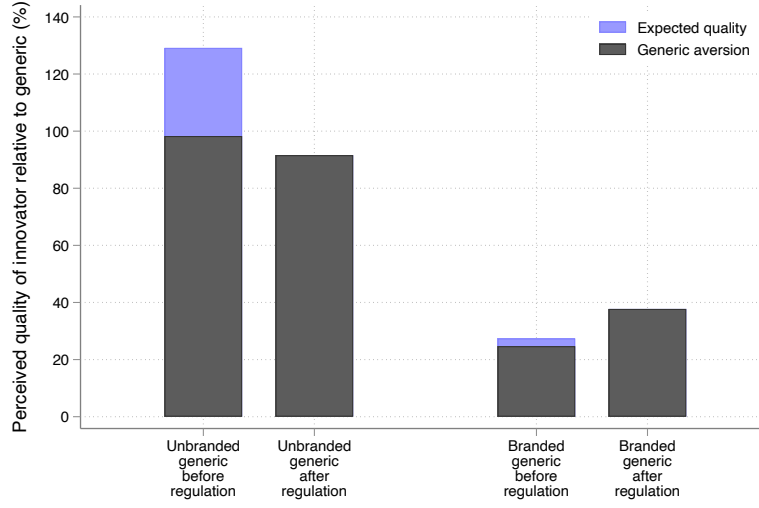
By combining demand estimates with estimates for the share of low-quality drugs from the supply model, we recover the parameters that govern aversion to generics. Specifically, by plugging our estimates into equation (10), we recover aversion to generics before and after the regulation as

$$\tau_{0,m}^k = \pi_H^k - \frac{v_{0,m}^k}{v_m^I}, \quad \tau_{1,m}^k = 1 - \frac{v_{1,m}^k}{v_m^I}, \quad (12)$$

with which we quantify the contributions of expected quality differences and aversion to generics to the lower valuation of generics before the reform, as well as the extent to which quality regulation impacted aversion.

Our estimates imply substantial aversion to unbranded generics and, to some extent, branded generics. Before quality regulation, aversion to generics was on average $\bar{\tau}_0^U = 0.98$ and $\bar{\tau}_0^B = 0.25$. These estimates are qualitatively in line with previous research showing that aversion to

Figure 5: Perceived quality gaps and quality regulation



Notes: This figure displays the perceived valuation of innovators relative to unbranded and branded generics. Each bar is decomposed into the components of perceived valuation gaps related to aversion to generics (gray) and drug quality (blue).

generics contributes to brand premiums (e.g., Bronnenberg *et al.*, 2015). The estimated magnitude of aversion to generics is particularly large for unbranded generics. These large differences in perceived valuation are necessary to rationalize the fact that innovator drugs and branded generics held large market shares despite charging much higher prices than unbranded generics.

Quality regulation had different effects on aversion to generics across segments. Aversion to unbranded generics decreased to $\hat{\tau}_1^U = 0.91$. This decrease helps rationalize that their perceived valuation increased by more than the increase in their average quality. This finding implies that, on top of the direct effects on drug quality, the regulation further decreased vertical differentiation by reducing aversion to unbranded generics. Conversely, aversion to branded generics increased after the policy, with $\hat{\tau}_1^B = 0.38$. This result helps rationalize that their valuation did not increase even though their average quality increased. This finding suggests that the policy revealed to consumers that branded generics are indeed generics and separated them more from innovator drugs relative to baseline. This may be related to the fact that all bioequivalent generics received the same distinctive yellow label, as shown in Figure A.1. This interpretation is consistent with the high market share of branded generics at baseline despite their high relative prices.

Figure 5 displays a decomposition of perceived innovator quality premiums.³⁵ Around 80%

³⁵We decompose differences in valuations between innovator and generic drugs as:

$$\hat{v}_m^I - \hat{v}_{t,m}^k = (\hat{v}_m^I - \hat{v}_{t,m}^k) + (\hat{v}_{t,m}^k - \hat{v}_{t,m}^k),$$

where $t = 0$ and $t = 1$ denote environments without and with quality regulation; $\hat{v}_m^I$ and $\hat{v}_{t,m}^k$ are the estimated valuations for drugs in market m among innovators and generics in segment k ; and $\hat{v}_{t,m}^k$ is the valuation of generics in segment k if

of the increase in the relative perceived valuation of unbranded generics was due to resolving asymmetric information, whereas the remainder was due to decreases in aversion to generics. These results are consistent with our survey evidence, which also points to large baseline gaps in perceived quality and their partial reduction due to the regulation.

7 Equilibrium Effects of Quality Regulation

Quality regulation affects market outcomes through four channels. First, it resolves asymmetric information by making drug quality observable to consumers, thereby increasing their willingness to pay for generics. Second, it changes the distribution of drug quality by mechanically removing low-quality drugs through a minimum quality standard. Third, it impacts consumer aversion to generics. Finally, it induces drug exit by imposing certification costs.

The model structure allows us to decompose the overall effects into these different channels. Specifically, let an equilibrium outcome be summarized by $y = y(E[\psi], \underline{\psi}, \tau, \kappa)$. In this expression, $E[\psi]$ captures consumer expectations of drug quality—namely, π and ψ at baseline and after quality regulation, respectively. Moreover, $\underline{\psi}$ is the minimum quality standard, which rises to ψ_H once the standard is in place. We decompose the effects of quality regulation on y as

$$\begin{aligned} \Delta y = & \underbrace{y(\psi, \psi_L, \tau_0, 0) - y(\pi_H, \psi_L, \tau_0, 0)}_{\Delta \text{ due to no asymmetric information}} + \underbrace{y(\psi, \psi_H, \tau_0, 0) - y(\psi, \psi_L, \tau_0, 0)}_{\Delta \text{ due to minimum quality standard}} \\ & + \underbrace{y(\psi, \psi_H, \tau_1, 0) - y(\psi, \psi_H, \tau_0, 0)}_{\Delta \text{ due to changes in aversion to generics}} + \underbrace{y(\psi, \psi_H, \tau_1, \kappa) - y(\psi, \psi_H, \tau_1, 0)}_{\Delta \text{ due to certification cost}}. \end{aligned} \quad (13)$$

We first simulate equilibrium outcomes without quality regulation, $y(\pi_H, \psi_L, \tau_0, 0)$. Using the resulting market structure as a baseline, we simulate equilibrium outcomes under each environment up to the full policy, $y(\psi, \psi_H, \tau_1, \kappa)$. See Appendix E.6 for simulation details.

7.1 Effects on Market Outcomes

We begin by simulating market outcomes for an environment without quality regulation, in which there is asymmetric information and the share of high-quality drugs is the estimated π_H^k , aversion to generics is given by $\tau_{t,m}^k = \tau_{0,m}^k$, and certification costs are $\kappa = 0$. The results for this baseline environment are shown in column 1 of Table 4-A. The simulated outcomes align with the main industry patterns documented in Section 2.4 regarding market structure, market shares, and prices across segments. Starting from this baseline, we compute each term in equation (13) to decompose

they were of high quality ($\pi_H^k = 1$) but at the estimated aversion to generics ($\tau_{t,m}^k = \hat{\tau}_{t,m}^k$). The first term captures the part of the gap due to aversion to generics, and the second term captures the part due to expected quality differences. With quality regulation, the second term is mechanically zero since there are no remaining differences in expected quality.

Table 4: Decomposition of the effects of quality regulation

		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
		Baseline	+Δ Due to no asymmetric information		+Δ Due to quality standard		+Δ Due to generic aversion		+Δ Due to certification cost	
	Segment	y	Δy	$\Delta \%y$	Δy	$\Delta \%y$	Δy	$\Delta \%y$	Δy	$\Delta \%y$
<i>A - Market outcomes</i>										
Number of firms	Innovator	0.86	-0.01	-0.63	-0.01	-0.63	-0.01	-0.30	-0.01	-0.30
	Branded	7.22	-0.10	-1.39	-0.30	-3.62	-0.51	-6.19	-0.91	-13.33
	Unbranded	6.20	0.43	7.43	-1.60	-19.88	-1.50	-17.79	-1.93	-25.71
Average price	Innovator	3.60	-0.04	-2.12	-0.06	-2.21	0.00	-0.59	0.03	0.97
	Branded	2.02	-0.00	-0.08	0.00	0.29	0.00	0.15	0.06	2.61
	Unbranded	0.57	0.02	4.61	0.03	8.80	0.04	9.76	0.07	23.11
Market share	Innovator	0.25	-0.03	-2.44	-0.03	-2.64	-0.01	-0.56	0.00	0.18
	Branded	0.48	-0.08	-5.54	-0.05	-2.97	-0.12	-8.03	-0.12	-7.77
	Unbranded	0.27	0.11	9.12	0.08	6.31	0.13	10.72	0.12	9.99
Share-weighted quality		0.92	0.08		0.08		0.08		0.08	
<i>B - Welfare effects (MM/year)</i>										
Compensating variation (A1)		1,112.95	47.41	4.26	48.20	4.33	45.80	4.12	43.10	3.87
Compensating variation (A2)		1,265.49	80.89	6.39	79.77	6.30	81.32	6.43	77.84	6.15
Variable profits		102.46	-2.52	-2.46	-3.13	-3.05	-1.02	-1.00	-0.54	-0.52
Certification costs		0.00	0.00	.	0.00	.	0.00	.	8.88	.
Fixed costs		27.59	0.54	1.95	-3.56	-12.90	-3.85	-13.95	-5.54	-20.08
Total profits		74.87	-3.05	-4.08	0.43	0.58	2.83	3.78	-3.87	-5.17
Welfare (A1)		1,187.82	44.35	3.73	48.63	4.09	48.63	4.09	39.23	3.30
Welfare (A2)		1,340.36	77.84	5.81	80.20	5.98	84.15	6.28	73.96	5.52

Notes: Column 1 shows baseline outcomes without the reform. The rest of the table decomposes the effect of quality regulation. Columns 2 and 3 show changes in outcomes relative to column 1 coming from resolving asymmetric information about drug quality, but without imposing a minimum quality standard. Columns 4 and 5 display changes relative to column 1 after imposing the minimum quality standard. Columns 6 and 7 display changes relative to column 1 after also accounting for changes to aversion to generics. Columns 8 and 9 display the changes relative to column 1 after also accounting for certification costs. This is the full effect of quality regulation. Compensating variation and total welfare are reported according to assumptions A1 and A2 in Section 7.2. To compute baseline consumer welfare, we calculate the compensating variation between the baseline equilibrium and an environment in which only the outside option is available. Fixed and certification costs are reported in annualized terms. Results for market outcomes report averages across markets. In particular, Δy denotes the average level change in outcome y relative to baseline across markets, and $\Delta \%y$ denotes the average percentage change in y relative to baseline across markets. Welfare results are aggregated across all markets.

the impacts of quality regulation.

In the first step, we isolate the effect of resolving asymmetric information about drug quality by setting $E[\psi_j] = \psi_j$. Columns 2 and 3 of Table 4-A show the results. Resolving asymmetric information leads to an increase in willingness to pay for high-quality generics. This demand shift is most substantial for unbranded generics, since their average quality at baseline is the lowest. Higher demand lures high-quality generic entry and induces low-quality generic exit, which brings the average quality of the generic drug pool close to that of the innovator.

In the second step, we add the effect of a minimum quality standard $\underline{\psi} = \psi_H$, by forcing the remaining low-quality drugs with $\psi_j = \psi_L$ to exit. Columns 4 and 5 of Table 4-A show the results. There is net exit among branded and unbranded generics. The number of firms in each segment

is 3.6 and 19.9% lower, respectively.³⁶ Coupled with drug exit, the increase in willingness to pay for generics induces price increases among generics. At this point, the market is characterized by high-quality drugs, higher prices, and a demand shift toward unbranded generics.

In the third step, we quantify the additional effects of decreasing aversion to generics. In particular, we set $\tau_{t,m}^k = \tau_{1,m}^k$. Columns 6 and 7 of Table 4-A show the changes in equilibrium outcomes from the effects of quality regulation on top of imposing a quality standard. Setting aversion to generics to its post-regulation level makes unbranded generics more competitive and branded generics less competitive, and results in the entry of the former and exit of the latter. Along the same line, unbranded generics gain substantial market share and increase their prices.

Finally, in the fourth step, we introduce the certification cost κ to obtain the full effect of quality regulation. Columns 8 and 9 of Table 4-A show the results. Adding certification costs reduces the number of branded and unbranded generics in the market, which renders the policy's negative effect on the total number of drugs in the market 41% larger. As a result, price increases among generics more than double relative to an environment without certification costs.

These are the full effects of quality regulation. Overall, our model predicts reductions in the number of drugs for all segments, coupled with price increases among generic drugs. The magnitudes of these simulated effects are close to those in our descriptive evidence. We find that resolving asymmetric information is the primary driver of the effects of quality regulation: Higher demand for unbranded generics leads to higher prices and net entry of drugs in this segment. The quality standard removes all remaining low-quality drugs from the market, thereby relaxing competition and increasing prices. Changes in aversion to generics amplify the substitution toward unbranded generics because of the increased aversion to branded generics coupled with a small decrease in aversion to unbranded generics. Finally, certification costs induce further exit, which amplifies the adverse effects on market structure and prices.³⁷

7.2 Welfare Analysis

Two main frictions in consumer choice shape the welfare effects of quality regulation. First, asymmetric information about product quality implies that consumers who purchase a non-bioequivalent generic would have been better off with a different product. Therefore, eliminating non-bioequivalent generics from the choice set may improve consumer welfare. Second, the policy

³⁶The net exit rate due to the quality standard is higher than the share of low-quality drugs among branded generics (3.6% as opposed to 3%), and lower than the share of low-quality drugs among unbranded generics (19.9% as opposed to 31%). Evaluating this policy without accounting for this margin would have led to incorrect conclusions about the distribution of drug quality. These results underscore the relevance of accounting for endogenous firm entry and exit responses to regulation, as recent work has also pointed out (Fan, 2013; Wollmann, 2018).

³⁷Note that whereas the decomposition relies on the assumed structure of the perceived value of drugs in equation (4)—how expected quality and aversion to generics enter utility along with their respective estimates—the overall effects are independent of those and only depend on demand estimates in equation (11).

may improve consumer choices by decreasing biases against generics (e.g., biased beliefs about drug quality). Biases decrease welfare by inducing suboptimal choices.

These demand-side frictions provide two reasons not to adopt a revealed preference approach to welfare analysis in this setting. More precisely, these frictions introduce two wedges between choice utility v_{jmt}^k and experienced utility $\tilde{v}_{jmt}^k$, which determine choices and welfare, respectively. First, choice utility is based on expected quality $E[\psi_j]$, whereas experienced utility is based on actual quality ψ_j . Second, choice utility is based on a degree of aversion to generics τ_m^k , which may be only partially welfare-relevant. We evaluate the welfare effects of the regulation under two different assumptions for experienced utility:

$$\text{A1: } \tilde{v}_{jmt}^k = \mu_m^I \cdot (\psi_j - \tau_{1,m}^k)$$

$$\text{A2: } \tilde{v}_{jmt}^k = \mu_m^I \cdot (\psi_j - 0),$$

both of which treat actual quality rather than expected quality as welfare-relevant, but take different stances for the component of aversion to generics that is welfare-relevant.³⁸

Under assumption A1, the welfare-relevant aversion to generics is what we estimated for the post-reform period, $\tau_{1,m}^k$. That is, consumers are not biased against generics after the reform, and hence their aversion to generics at that point reflects lower experienced utility (in the pre- and post-reform environments). This assumption corresponds to an environment where the regulation removes all the welfare-irrelevant components of aversion to generics.³⁹ Alternatively, under assumption A2, none of the estimated aversion to generics is welfare-relevant, so bioequivalent generics provide the same benefits as innovator drugs. Assumption A1 would be preferred over A2 if, for example, some quality differences remain after bioequivalence requirements are in place.⁴⁰ Conversely, assumption A2 would be preferred over A1 if consumers still hold biased beliefs about quality after the regulation, or if demand is still affected by manufacturers' detailing efforts.

We calculate the total welfare effect of counterfactual policies as

$$\Delta W = \sum_{i,m} CV_{im} + \sum_{j,m} \Delta \tilde{\Pi}_{jm} - \sum_{j,m} \Delta F_{jm} - \sum_{j,m} \Delta \kappa_{jm},$$

which includes consumer compensating variation CV_{im} , the change in firm variable profits $\Delta \tilde{\Pi}_{jm}$,

³⁸See, e.g., Handel (2013) for a similar approach in the context of consumer inertia in insurance choice, and Grennan *et al.* (2025) for a welfare analysis of the effects of physician payments on demand for drugs.

³⁹In other words, under this assumption, quality regulation turns consumers into "experts". This method for uncovering the welfare-relevant part of aversion is similar to that in Bronnenberg *et al.* (2015), who use demand from experts, with the added assumption that individuals become experts after quality regulation.

⁴⁰For example, Ishitani (2025) documents high recall rates among bioequivalent generics in the U.S., associated with a lack of compliance with manufacturing practices, cross-contamination, among other reasons.

changes in the amount of fixed costs incurred by entrants ΔF_{jm} , and changes in the amount of certification costs paid by generics $\Delta \kappa_{jm}$. Appendix E.4 provides more details.

The results of our welfare analysis of quality regulation are shown in Table 4-B. Overall, the regulation increases welfare by between \$39 and \$74 million per year, depending on the welfare metric used. As expected, welfare increases more under assumption A2, under which aversion to generics is not welfare-relevant, since there is a net increase in the market share of generics. Increases in consumer welfare drive the bulk of the overall welfare increase. On the other hand, firms are worse off under the regulation, since their variable profits decrease by \$0.5 million and they must incur \$8.9 million in certification costs—albeit lower entry saves \$5.5 million in fixed costs. Taken together, these results suggest that the gains from increased drug quality and the reduction in incurred entry costs more than offset the losses from the lower variety and higher prices associated with the adverse competitive effects of certification costs.⁴¹

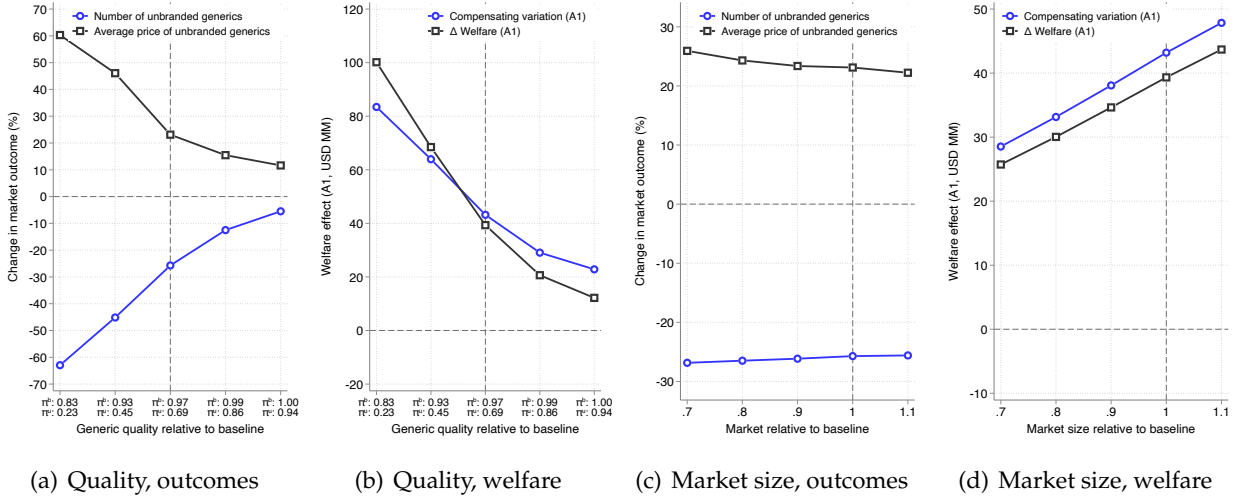
Resolving asymmetric information is the main driver of the positive welfare impacts of quality regulation. Using the decomposition in equation (13), we find that this component of the policy alone increases welfare by between \$44 and \$78 million. Accounting for the impacts of the quality standard and shifts in aversion to generics increases these welfare effects by around 10%. Conversely, the negative impacts of certification costs on market structure and competition reduce welfare, thereby limiting the overall welfare gains from the regulation.

Connection to descriptive patterns. It is worth discussing how the findings from our welfare analysis relate to the main descriptive patterns in market outcomes. Quality regulation may impact consumer welfare by affecting market outcomes through three channels: (i) the exit of low-quality generics in response to the quality standard, which implies consumers cannot obtain a low-quality drug, (ii) changes in market shares, and (iii) changes in prices.

Regarding the first channel, the data show that unbranded generics have a baseline market share as high as 26%, even though we estimate that 31% of these were non-bioequivalent. The quality standard eliminates the welfare losses associated with the purchase of low-quality generics. Through the lens of the model, we infer that the welfare loss from buying a low-quality generic instead of a bioequivalent generic is substantial, as reflected by the low market share of unbranded generics despite their lower prices. Regarding the second channel, we find that quality regulation induces net substitution toward high-quality unbranded generics. This shift leads to consumer welfare gains insofar as lower expenditures more than compensate for losses due to aversion to generics. Regarding the third channel, quality regulation led to higher prices for unbranded

⁴¹The results of our welfare analysis depend on the value assigned to low-quality drugs. We note that our main demand model specification normalizes this value to zero. In Appendix E.7, we discuss this normalization and show that our results are robust to other plausible normalizations. In particular, Figure A.6 shows that the impacts of quality regulation on consumer and total welfare remain positive across a broad range of assumptions about the value of low-quality drugs under both assumptions A1 and A2.

Figure 6: Quality regulation under counterfactual baseline quality and market size



Notes: Each dot represents the full effect of quality regulation—relative to a baseline without quality regulation—for a range of industry primitives. For each configuration, we simulate the model for both environments. Panels (a) and (b) report results for a range of distributions of baseline quality of generics, where we vary π_H^k . Panels (c) and (d) report results for a range of market sizes relative to baseline. Panels (a) and (c) display effects on the number (blue) and price (black) of unbranded generics as a subset of market outcomes. Panels (b) and (d) display effects on consumer (blue) and total (black) welfare as a subset of welfare outcomes. Vertical dotted lines indicate baseline industry primitives—namely, estimates of average drug quality by segment and observed market size.

generics, as documented in Section 3.2. While these price increases are detrimental to welfare per se, our model allows us to establish that they are not large enough to outweigh the gains from improving the pool of unbranded generics.

7.3 Drivers of the Equilibrium Effects of Quality Regulation

Our welfare analysis shows that quality regulation was welfare-enhancing in aggregate. However, the underlying trade-offs suggest that this regulation could also have detrimental effects on welfare. Indeed, quality regulation leads to lower welfare for 22% of the markets in our sample. Market size is a key correlate of this heterogeneity: this fraction is 44% in markets below the median size, but only 2% for those above it. This pattern is consistent with the descriptive evidence in Section 3.2, which showed stronger impacts of the regulation on market structure among smaller markets.

We further explore the determinants of the effects of quality regulation by implementing the same analysis under alternative market configurations. In particular, we explore changes along two key dimensions: the distribution of drug quality at baseline and market size. In practice, we simulate baseline outcomes without quality regulation across a range of alternative configurations and then evaluate the impacts of quality regulation starting from each such baseline. Figure 6 displays the results of this analysis. First, Figures 6-a and 6-b show that quality regulation would have substantially stronger welfare effects in markets with lower baseline expected quality, in

which resolving asymmetric information and removing low-quality drugs from the market would be most impactful. Second, Figures 6-c and 6-d show that quality regulation would be more welfare-enhancing for large markets, among which certification costs have less incidence.

8 Policy Counterfactuals

Certification cost subsidies. Certification costs limit the welfare gains from quality regulation by inducing exit and price increases. We study whether certification subsidies could limit these unintended consequences. Table 5-A shows results for an environment in which certification costs are subsidized at 50%. Compared with the environment that includes quality regulation but no subsidies in columns 8 and 9 of Table 4, this equilibrium features more firms and lower prices. Fully eliminating certification costs further limits both drug exit and price increases.

Subsidizing certification would be more impactful in smaller markets, consistent with the results in Section 7.3. In particular, when firms incur the full certification cost, quality regulation leads to decreases in the number of firms by 29.7 and 14.4% in small and large markets, respectively. When subsidizing certification costs, these adverse effects are 14 and 3.2 percentage points lower in small and large markets, respectively. Similarly, the positive impact of quality regulation on the average prices of unbranded generics is 4.6 percentage points lower with certification cost subsidies in smaller markets, as opposed to only 0.2 percentage points in large markets.⁴² These results suggest that subsidizing certification may enhance the impacts of quality regulation—particularly among smaller markets—and support the case for targeted subsidies.

Quality disclosure. An alternative to a minimum quality standard is quality disclosure. This policy involves informing consumers about the quality of each product while still allowing products that fall below the quality standard to be sold. To the extent that these products compete with high-quality products, keeping them in the market may reduce the adverse competitive effects associated with minimum quality standards, while still resolving asymmetric information. We compare these policies by implementing a disclosure policy in our model whereby firms must pay the same certification cost κ to verify their quality, but can stay in the market even if their drug is not bioequivalent.⁴³ After a firm tests for bioequivalence, its quality ψ_j becomes observable.

Quality disclosure outperforms minimum quality standards in terms of market outcomes, as shown in Table 5-B. Because low-quality firms are allowed to sell under quality disclosure, the number of firms decreases less than under quality regulation (see columns 8 and 9 of Table 4). The number of unbranded generics decreases by only 1%, driven by increased entry of high-quality

⁴²To further explore this comparison, Figure A.7 shows the effects of quality regulation with and without certification subsidies across markets. While quality regulation decreases the number of firms and increases unbranded generic prices in most markets, these effects are weaker with certification subsidies, particularly among smaller markets.

⁴³In terms of the effect of disclosure on aversion to generics, we assume that once the policy is in place, consumers have aversion $\tau_{1,m}^k$ to all drugs in a segment, regardless of their quality.

Table 5: Counterfactual analysis: Certification costs, quality disclosure, and aversion to generics

		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
		A - Role of certification costs					B - Disclosure		C - The role of aversion to generics					
		Amount of subsidy					Quality disclosure		Share of aversion to generics corrected by regulation					
		50%					100%		s = 0%					
		Δy	$\Delta\%y$	Δy	$\Delta\%y$	Δy	$\Delta\%y$	Δy	$\Delta\%y$	Δy	$\Delta\%y$	Δy	$\Delta\%y$	Δy
Baseline									s = 50%					
Segment		y						s = 100%						
Market outcomes														
Number of firms	Innovator	0.86	-0.01	-0.30	-0.01	-0.30	-0.01	-0.30	-0.01	-0.63	-0.04	-2.16	-0.13	-6.64
	Branded	7.22	-0.75	-10.46	-0.51	-6.17	-0.73	-11.38	-0.70	-10.78	-0.79	-12.10	-1.10	-15.64
	Unbranded	6.20	-1.72	-22.12	-1.49	-17.77	-0.03	-1.03	-2.05	-27.91	-1.49	-16.66	-1.12	-8.62
Average price	Innovator	3.60	0.01	0.06	0.00	-0.59	0.02	0.62	-0.03	-0.71	-0.17	-6.97	-0.34	-9.53
	Branded	2.02	0.04	1.61	0.00	0.15	0.05	2.05	0.06	3.08	0.05	3.33	0.05	2.95
	Unbranded	0.57	0.06	14.91	0.04	9.77	0.05	13.62	0.06	20.85	0.05	17.71	0.06	18.46
Market share	Innovator	0.25	-0.00	-0.21	-0.01	-0.56	0.00	0.10	-0.02	-1.92	-0.13	-10.07	-0.21	-16.41
	Branded	0.48	-0.12	-7.90	-0.12	-8.04	-0.12	-8.18	-0.04	-2.77	-0.14	-8.37	-0.33	-19.46
	Unbranded	0.27	0.12	10.38	0.13	10.73	0.12	10.29	0.06	5.68	0.27	24.28	0.54	49.79
Share-weighted quality		0.92	0.08	0.08	0.08	0.08	0.08	0.08	0.08	0.08	0.08	0.08	0.08	0.08
Welfare effects (MM/year)														
Compensating variation (A1)		1,112.95	44.63	4.01	46.04	4.14	43.44	3.90	45.36	4.08	55.03	4.94	56.61	5.09
Compensating variation (A2)		1,265.49	79.91	6.32	81.73	6.46	78.18	6.18	75.96	6.00	73.33	5.79	64.18	5.07
Variable profits		102.46	-0.78	-0.76	-1.02	-0.99	-0.59	-0.57	-2.64	-2.57	-9.54	-9.31	-11.49	-11.21
Certification costs		0.00	10.44	10.89	10.89	10.78	10.78	10.78	8.93	9.38	9.38	9.49	9.49	9.49
Cost of public funds (20%)		0.00	1.04	2.18	2.18	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Fixed costs		27.59	-4.80	-17.41	-3.84	-13.92	-1.74	-6.31	-5.27	-19.09	-4.47	-16.20	-4.51	-16.36
Total profits		74.87	-6.42	-8.57	-8.07	-10.78	-9.62	-12.85	-6.30	-8.42	-14.45	-19.30	-16.46	-21.99
Welfare (A1)		1,187.82	37.17	3.13	35.79	3.01	33.82	2.85	39.06	3.29	40.58	3.42	40.15	3.38
Welfare (A2)		1,340.36	72.45	5.41	71.48	5.33	68.56	5.11	69.66	5.20	58.88	4.39	47.72	3.56

Notes: Column 1 shows outcomes in the baseline without the reform. Panel A displays results for equilibria under regulation with certification costs subsidies. Columns 2 and 3 display the changes relative to column 1 under a 50% subsidy. Columns 4 and 5 show changes relative to column 1 under a full subsidy. Panel C displays results for equilibria under quality disclosure relative to column 1. Panel C describes how aversion to generics mediates the effects of quality regulation on market outcomes. Columns 8-13 display the effect of the regulation on market outcomes relative to baseline for different degrees of correction of aversion to generics. We define aversion to generics as a linear combination of the baseline aversion to generics and the aversion of a consumer with no aversion toward generics. Formally, $\tau_{j,m}^k(s) = (1-s)\tau_{j,m}^k$, where s measures the effect of quality regulation on aversion to generics. Compensating variation and total welfare are reported according to assumptions A1 and A2 in Section 7.2. To compute baseline consumer welfare, we calculate the compensating variation between the baseline equilibrium and an environment in which only the outside option is available. Fixed and certification costs are reported in annualized terms. Results for market outcomes report averages across markets. In particular, Δy denotes the average level change in outcome y relative to baseline across markets, and $\Delta\%y$ denotes the average percentage change in y relative to baseline across markets. Welfare results are aggregated across all markets.

drugs and reduced exit of low-quality drugs compared to under a minimum quality standard. This occurs because low-quality incumbents are not forced to exit and must pay only the certification cost but not the entry cost to remain in the market, giving them an advantage over potential entrants and limiting turnover.⁴⁴ The weaker impact of quality disclosure on market structure results in smaller price increases compared to minimum quality standards.

Quality disclosure has a slightly stronger effect on consumer welfare than minimum quality standards. Like minimum quality standards, quality disclosure eliminates the welfare losses from consumers who buy a low-quality drug by resolving asymmetric information and has less adverse competitive effects. However, minimum quality standards dominate in terms of total welfare. By not allowing low-quality drugs to be sold, this policy reduces the total fixed and certification costs incurred and leads to higher overall welfare despite the weaker competition.

Lowering aversion to generics. Quality regulation aims to foster the use of generics by increasing their expected quality as well as decreasing aversion. Our estimates reveal baseline aversion to unbranded generics along with policy impacts on this margin. However, policymakers could adopt policies complementary to quality regulation that target aversion directly, such as information campaigns, regulating prescription behavior, or generic substitution, among others. We study how the impacts of quality regulation depend on its effects on aversion to generics by simulating the regulation for a range of post-reform aversion levels defined by $\tilde{\tau}_{1,m}^k(s) = (1 - s)\hat{\tau}_{0,m}^k$, where s is the share of aversion toward generics corrected by the policy.

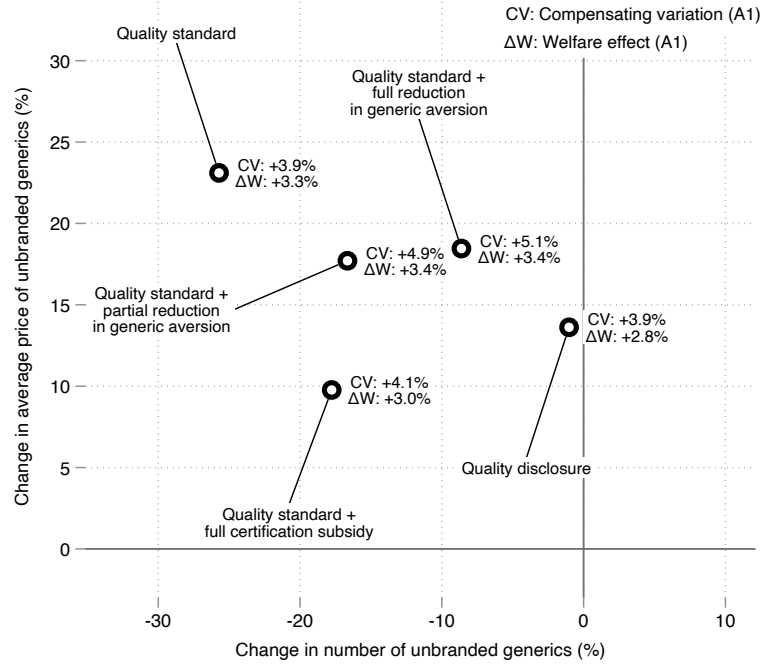
Whether quality regulation affects aversion to generics is an important determinant of its effects on market outcomes. Table 5-C presents the results. When quality regulation has no effect on aversion to generics, it has a limited effect on vertical differentiation and competition. These effects become stronger when the regulation reduces aversion to generics more substantially, as illustrated on the right of Table 5-C. As aversion to unbranded generics decreases, competition between innovators and branded generics intensifies. This leads to increased exit of branded generics alongside stronger price reductions for innovator drugs.

The welfare effects of quality regulation depend on how it influences perceived quality and the extent to which choice and experienced utility become more aligned. Modest reductions in generic aversion increase consumer welfare by improving the quality of decision-making. However, when these reductions are larger, competition becomes intense enough that the adverse effects of drug exit may outweigh those benefits. This latter mechanism dominates in our analysis, leading to increasingly smaller effects on consumer welfare under assumption A2.⁴⁵ Moreover, quality

⁴⁴Figure A.8 shows how quality regulation and disclosure impact the variable profits of low- and high-quality firms. While both firm types are equally profitable at baseline, they separate under both policies since quality can be priced once asymmetric information is resolved. Many low-quality firms remain profitable, but they operate only under disclosure.

⁴⁵We focus on assumption A2 because reducing aversion to generics has a mechanical effect on consumer welfare

Figure 7: Comparison of counterfactual policies



Notes: Each dot represents the effect of a counterfactual policy—relative to a baseline without quality regulation—for the number of unbranded generics (x-axis) and their average price (y-axis). Five policies are considered: (i) baseline quality regulation as a minimum quality standard in columns 8 and 9 of Table 4, (ii) quality regulation with fully subsidized certification costs in columns 4 and 5 of Table 5-A, (iii) quality disclosure in columns 6 and 7 of Table 5-B, (iv) quality regulation with partial reduction in aversion to generics in columns 10 and 11 of Table 5-C, and (v) quality regulation with a full reduction in aversion to generics in columns 12 and 13 of Table 5-C. For each policy, the figure also reports the compensating variation and welfare effect under assumption A1.

regulation yields smaller overall welfare gains when it has a stronger effect on aversion to generics, partly driven by reduced profits due to more intense competition by generics. These results suggest that complementary strategies aimed at closing perceived quality gaps across segments may influence the overall effectiveness of quality regulation.

Comparison of counterfactual policies. In Figure 7, we compare the policies we study in terms of their effects on market outcomes and welfare. A simple quality standard has the strongest adverse effect on both the number of unbranded generics and their prices. This partially explains why, in most cases, the other policies generate greater consumer welfare gains. Still, given that drug exit saves on certification and fixed costs, the simple quality standard yields among the largest total welfare gains among these policies.⁴⁶ In addition, quality disclosure is the policy under which

under A1, given it is a function of $\tau_{1,m}^k$.

⁴⁶This result is partly driven by the strong within-segment substitution patterns we estimate. A market with more substitutable products is more prone to inefficiently high entry (Mankiw and Whinston, 1986). In a setting with weaker substitutability, quality standards would be less likely to outperform quality disclosure in terms of welfare effects.

the number of unbranded generics decreases the least. While this helps limit the price effects of the regulation, the policy is outperformed in terms of welfare because some low-quality drugs remain in the market, which requires sinking additional fixed and certification costs. These results suggest that quality standards induce a stronger and better selection of products in the market. Importantly, we note that all of these policy designs increase welfare, as their primary impact is to resolve asymmetric information and improve the quality of the pool of drugs in the market.

9 Conclusion

Quality regulation in markets characterized by asymmetric information can improve welfare through several mechanisms: ensuring product quality, enhancing consumers' quality perceptions, and promoting price competition by reducing vertical differentiation. However, costly compliance with quality standards may induce firm exit and thus harm price competition. We study these trade-offs in the context of pharmaceutical markets, in which drug quality and asymmetric information are primary concerns. To this end, we leverage the introduction of bioequivalence requirements—the most common quality standard for generics—in the Chilean pharmaceutical market. Combining descriptive evidence with a structural model of the market, we find that quality regulation improved welfare despite its adverse effects on market structure and prices, and that resolving asymmetric information was the primary driver of these welfare gains.

Our analysis offers broader insights for the design of quality regulation. First, this regulation is more likely to improve welfare when baseline product quality is relatively low. Second, quality regulation may be combined with complementary policies to enhance its impact by influencing quality perceptions beyond resolving asymmetric information. Third, compliance costs are a key driver of the unintended consequences of quality regulation on market outcomes. Subsidizing certification can therefore serve as a powerful tool to mitigate these adverse effects, particularly in smaller markets. Finally, minimum quality standards may lead to a stronger and better product selection in the market than quality disclosure, but the latter may still yield greater consumer welfare by providing broader choice. Although our quantitative results are specific to our context, our model offers a rich framework for analyzing the design of quality regulation in other settings.

Although this paper addresses several aspects of how quality regulation affects equilibrium outcomes, certain questions remain unresolved. First, we document substantial aversion to generics, even after the implementation of quality regulation. Understanding the drivers of this aversion would shed light on complementary policies aimed at improving market outcomes. Second, we do not account for firm responses through quality investments. While this is a plausible assumption in our context, long-run evaluations of quality regulation—particularly where quality is more easily adjustable—should incorporate endogenous quality adjustments. Finally, assessing the long-run effects of quality regulation on health outcomes would be a natural complement to our analysis.

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